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Intelligent Scholarship Application, Ranking, and Early-Warning Monitoring System HEIs



AN UNDERGRADUATE CAPSTONE PROJECT

Presented to the

Faculty of College of Computer Studies

St. Peter's College

Iligan City

In Partial Fulfillment

of the Requirements for the Degree of

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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1.1 Background of the Study

Digital management systems have been implemented in higher education institutions to enhance their service delivery, improve their operational efficiency, and provide data-driven decision-making (World Bank, 2020). Some administrative functions, such as record keeping, can be automated using web-based platforms which enables institutions to keep records more accurately and efficiently. Scholarship administration is among the key functions in higher education where students are given financial assistance to those who are meritorious and deserving. As the number of scholarship programmes increases, however, institutions are facing increasing difficulties in handling applications, confirming their eligibility and checking on the on-going compliance of scholars. The challenges are to do with manual and fragmented administrative processes which lowers the efficiency, transparency and uniformity of scholarship distribution (Al-Ayyubi & Maulana, 2023). Scholarship coordinators at St. Peter's College continue to do manual verification of enrolment by looking at the applicants' Certificates of Registration and student IDs. Additionally, they personally monitor requirements that are not completed and signatures required to disburse scholarships, further delaying the timely submission of scholarship paperwork to CHED. Previous research has demonstrated the effectiveness of web-based systems to improve data processing and organization (Al-Ayyubi & Maulana, 2023; Ablen et al., 2025), and the use of AI and machine-learning for objective, criteria-based applicant ranking (Fajardo et al., 2024). Even with these changes, St. Peter's College continues to depend on manual verification, which makes it hard to keep track of applicants and monitor scholars. This is the gap and what a system must provide to allow the online application for scholarship, rank the applicants and constantly monitor the scholarship recipients.

A significant number of institutions still rely on manual and paper-based approaches for their scholarship processes, frequently characterized by inaccuracies in the transcription of applications, delays in processing, and restricted transparency about the academic status of the scholarship recipients (Al-Ayyubi & Maulana, 2023; Bañeres et al., 2023). The following issues were found at St. Peter's College during the observations and interviews with the Scholarship Coordinator, where it has been observed that about 1,800 scholarship records that were approved by the Commission on Higher Education (CHED) for Academic Year 2025-2026 were still pending manual

validation prior to payment of benefits. It took the coordinator approximately 1 week to validate the records of one scholar and when performed over the many scholars validated in an academic year this can become burdensome. The coordinator manually verified scholars' Certificate of Registration (COR) with the Registrar's Office to ensure that enrollment continued. Personal calls were also made by the coordinator to scholars on their registered mobile phone to get required documents like a valid school ID, COR, and signature forms. This resulted in three persistent issues. First, the final list of beneficiaries could be inaccurate because some approved scholars could not be enrolled or might not be attending classes. Second, the mobile numbers of some scholars had changed, which affected about 10 percent of the scholars and resulted in delay in submission of missing requirements for CHED approval, according to the coordinator. Third, gathering of documents and signatures was more challenging and therefore missed deadlines imposed by CHED and other funding agencies. In-line with the above, the following issues are also reported in other manual and decentralised scholarship systems in literature, especially in regard to record management, communication, and monitoring (Ablen et al., 2025; Remegio & Casquejo, 2025). Moreover, they do not have an automatic academic monitoring system in place, making it hard to track scholars who might not meet retention standards. Early-warning systems have been proven to be effective in identifying at-risk students and facilitating timely interventions (Hu et al., 2014; Plak et al., 2022; Bañeres et al., 2023; Liu & Liu, 2025).

To address these concerns, this research suggests the creation of an Intelligent Scholarship Application, Ranking and Early-Warning Monitoring System for St. Peter's College. The system will enable on-line scholarship applications, centralized administration and predictive academic monitoring. Since the actual registrar system is unavailable, the proposed system enables registration of students as both freshman and alumnus applicants and verifies their identity and enrollment with a simulated Registrar Information System (RIS) that contains student information and grades for testing purposes. Once verified, students can check out the scholarship options, apply online, track applications and receive announcements and notifications of results. This eliminates reliance on paper applications and ensures that administrators have a single point to access, review, validate and control applicant data. The system has a machine learning based ranking module as well as an early-warning monitoring

module. Family income data from filed Income Tax Returns (ITR) is used to rank the applicants using the Random Forest Classifier, with the intention of ranking those with low family income and high financial requirements. This will help to make sure that the limited places for scholarships go to the most deserving applicants. The Early Warning Monitoring Feature (EWMF) is a simulated grades database that is used to monitor scholars' academic progress and automatically alerts students who are not making grade 93, which means they are at risk of losing their scholarship. They are then notified to the student and faculty members to support timely interventions. This is aligned with previous research that has indicated that early-warning systems can identify at-risk students and enhance academic outcomes by providing timely support (Hu et al., 2014; Bañeres et al., 2023; Liu & Liu, 2025). The overall goal of the proposed system is to enhance the process of administering scholarships at St. Peter's College in terms of making the application, ranking, and monitoring of the scholarships more efficient, accurate and uniform.

Unlike existing scholarship management systems, this study integrates the application process, application verification, academic monitoring and workflow management into a single platform using machine learning approach. So far, the studies have concentrated on one of the various stages of the scholarship process, with Baharuddin et al. (2024) presenting a decision support system to select the scholarships as well as Berido et al. (2023) presenting an automation-based scholarship process, but none of them monitored the scholar(s) after the selection process. Similarly, other researchers have also used machine learning methods like neural networks, TOPSIS, and different classification methods to evaluate scholarships and select candidates (Alfina et al., 2022; Cahyani et al., 2022; Chaudhari et al., 2022) but most of their works only focused on ranking the applicants during the selection process. The proposed system, however, includes not only the application and the verification phase but also the ranking of applications and the approval process of the applications, as well as the tracking of the recipients in real-time, meaning that it addresses the entire process of administration of the scholarship compared to just one of the stages of the application. The system will use family income declared in the income tax return as the most important feature to calculate the financial need of the applicants and will rank the applicants according to the financial need using random forest classifier. This algorithm is appropriate, because

the ensemble learning is based on the combination of multiple decision trees that allows it to deal with nonlinear relationships and interactions between variables, reduces the possibility of overfitting, and increases the stability and accuracy of the predictions in classification problems (Buslim et al., 2023; Kanwal et al., 2024; Paramesha et al., 2024). The proposed system also facilitates the objective and consistent selection of the scholars, without having to rely on personal judgment, and avoids any chance of biased and subjective selections. It also enables the continuous monitoring of the eligibility scholars, thus helping in the selection of scholar candidates as required.

To ensure accessibility, efficiency, and scalability, the system will be implemented as a web-based platform. This situation is appropriate because it does not need the local installation of the software, and can be accessed from any device that connects to the internet, like a laptop, tablet or a smartphone. This is particularly useful in the Philippines as the use of the internet keeps increasing and mobile telephones are the key mode of communication for many of our citizens (Philippine Statistics Authority [PSA] & Philippine Department of Information and Communications Technology [DICT], 2025). In the context of higher education institutions, a web-based system is also beneficial since it provides access for multiple end users, such as students, administrators, faculty members, guidance personnel and CHED staff, without the need of installing software or making settings on the devices. The front-end interface will be developed using React.js and Vite, the back-end processing will be done using Python (FastAPI or Flask), and a centralized database management system will be Supabase (PostgreSQL). The centralized approach allows for storing and synchronizing scholarship records, application information, ranking outcomes, and monitoring updates from a single place, minimizing inconsistencies that often occur due to spreadsheet or paper processing. It also enhances maintainability, real-time updating capability, and better managing of scholarship information for various user types. Past research indicates that web-based information systems enhance the management of data in a centralized way, access to information, and operational efficiency in educational environments (Al-Ayyubi & Maulana, 2023; Baharuddin et al., 2024). In this study, the aforementioned web-based platform enables students to submit applications on-line, check status of applications, administrators to check and

rank applicants, and the faculty/guidance staff to keep track of scholars flagged by the early-warning feature.

Multiple stakeholders are expected to benefit from the system, as its features are designed to support their specific roles and responsibilities. The online application, document upload, tracking and notification feature offers a more efficient, transparent and paperless journey for students to submit their scholarship applications, track their status and receive notifications in real time (Al-Ayyubi & Maulana, 2023; Ablen et al., 2025). The centralized database, applicant validation tools and the random forest based ranking module allow administrators, especially the Scholarship Coordinator, to lower the administrative workload and to help in the process of making more objective and consistent decisions since many records are processed every academic year, reducing repetitive manual efforts to verify or review the records on a case by case basis (Ablen et al., 2025; Remegio & Casquejo, 2025). The Early Warning Monitoring Feature allows for the early detection of students who might not be able to complete the program in a timely manner, enabling faculty and guidance staff to intervene before a student has completely failed to fulfill program completion requirements (Hu et al., 2014; Plak et al., 2022; Bañeres et al., 2023). The system helps CHED staff to have more organized rankings on applicants, validation information, and approved data that aid in a more structured and evidence-based scholarship review process (Buslim et al., 2023; Fajardo et al., 2024; Kanwal et al., 2024). The system is overall expected to increase the efficiency, transparency and data-oriented decision making processes in the administration of scholarships at St. Peter's College.

Overall, the system is expected to improve the efficiency of scholarship processing, the consistency of applicant evaluation, the effectiveness of academic intervention, and the transparency and reliability of scholarship administration. The system is designed to automate the verification of all applicants, their ranking and notification of the application status, thereby saving the Scholarship Coordinator from doing tedious manual work, which currently takes around one week per scholar, to just a few minutes in the automated mode by checking the applications against the simulated Registrar Information System. It is also expected that the use of the Random Forest Classifier will also enhance the consistency and objectivity of the ranking of scholarships as it assesses the applicant using the same financial-need criteria, especially family income based on the applicant's submitted Income Tax Return (ITR)

which will reduce the variability in the manual assessment. Also, the Early Warning Monitoring Feature will help to ensure that students are identified as needing early academic intervention, with the ability to automatically alert students to their falling below the required Grade Point Average (GPA) maintaining grade. This early detection enables faculty and coordinators to support scholars before they do not meet retention requirements, potentially improving their academic performance and scholar retention. The proposed system should help to enhance the data-driven decision making, boost the efficiency of the scholarship processing, and the monitoring and management of the scholarship recipients during the scholarship process.

1.2 Conceptual Framework

A conceptual framework is a graphical model of the key elements and relationships that are used to guide the development and assessment of a study. The conceptual framework for the proposed Intelligent Scholarship Application, Ranking, and Early-Warning Monitoring System (ISEWM) is presented as Input-Process-Output (IPO) model in this study. The framework shows how student- and scholarship-related data is collected, transformed, and used to produce outputs that can be used for validation, ranking, CHED approval, scholarship management and early-warning monitoring.

Input. Input component is the data needed for the system for applying for and monitoring scholarships. These are the list of Scholarship Programs, Application Forms, Student Information (Student ID, name, sex, birthdate, program, year level), financial Information (Annual Family Income, as reflected on the Income Tax Return (ITR)), supporting documents (Cert of Registration, valid ID, Signature Form) and academic Information (Senior High School General Weighted Average (GWA) for freshmen applicants and College Grade Point Average (GPA) for continuing and alumni applicants). To validate the system, a simulated registrar enrollment and grades database has also been used to check that a Student ID or name is that of a registered student, as access to the actual school's registrar's information system is limited by data confidentiality policies.

Process. Process component – procedures which convert the input collected to meaningful output. The system starts with user registration and management, then validates the student information, enrollment information and grades from the registrar

database. Applications for scholarships are then made and assessed for eligibility. The financial need and other relevant information available to the applicant is used to rank the applicants using the Random Forest Classifier. The rankings are sent to CHED for review and qualification of beneficiaries and the qualified beneficiaries are returned to the administrator. The system also tracks scholars' academic achievements and issues an early-warning alert if the scholar's grade drops below the required 93 percent maintenance of grade.

Output. The output component is set of the results generated by the system. These include the ranked list of scholarship applicants, application status notification (Pending or Approved), validated and approved beneficiaries, notifications and announcements of scholarship updates, and early-warning alerts for students who are at risk of losing eligibility for a scholarship. The system also enables administrators to check the ranking list or export it for submission to CHED and to validate student enrolment data based on integrated registrar data without having to manually review the Certificate of Registration. After CHED completes the list of "approved beneficiaries," it will be sent back to administrator for student notification and scholarship announcement. Admin can view ,export the support documents and academics records for emergency purpose for requirements student cant comply the documents physical admin can only export and submit it also this have a valid id with signature uploaded to students in their profile manage and if CHED needed requirements like this can export print it and submit also for verify manual if needed.

Feedback. Feedback is provided so that results of the system can be used as updated input to subsequent scholarship cycles. Academic standing, approval outcomes and reapplication data can be incorporated into the system to aid continuous improvement and ensure fairness in the selection of scholarships.

Further described in Chapter 3, these operational processes were designed and refined in an iterative process model, by making features better and better over the course of several development cycles.

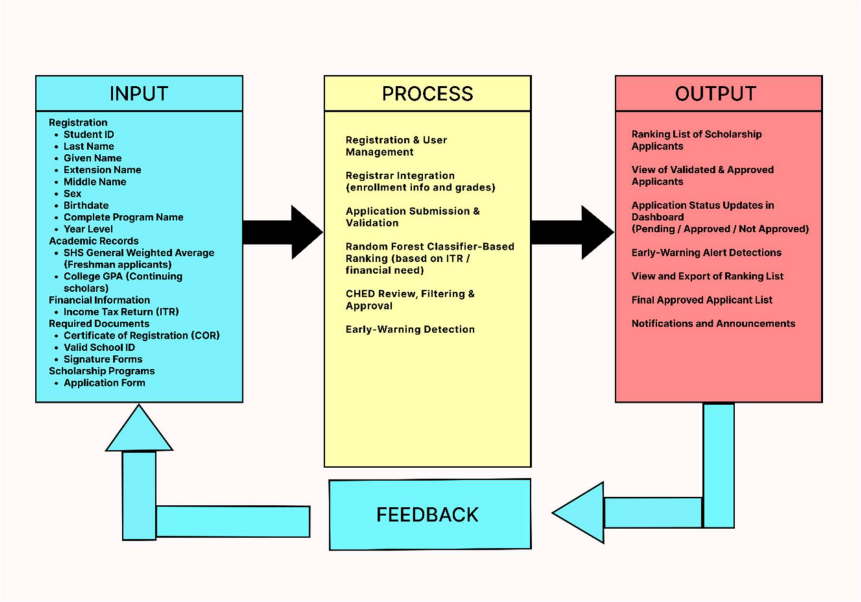


Figure 1. Conceptual Diagram Using the IPO Model

This figure 1 shows IPO the system starts with a student registration form, academic records, financial information, supporting documents, information about the scholarship program and simulated registrar data to be validated. The inputs are then fed into user management, record check and validation, application submission, applicant ranking with Random Forest Classifier, CHED review and approval and early-warning monitoring for scholars not achieving maintaining grade. The outputs are the ranked list of applicants, validated and approved beneficiaries, application status update, early-warning alerts and notifications and announcements. The feedback component makes it possible to apply the updated results and records in future cycles of scholarships, hence enhancing the efficiency, accuracy and fairness of the system.

1.3 System Framework

This section presents the framework of the proposed Intelligent Scholarship Application, Ranking, and Early-Warning Monitoring System with a three-tier client-server architecture: Presentation Layer, Application Layer and Data Layer. This architecture helps achieve modularity, scalability, maintainability, and secure management of scholarship data, by breaking it into three components: user interface, system logic, and data storage.

Developed with React.js and Vite, the Presentation Layer represents the user interface of the system. This layer allows students to sign up for scholarships, apply for scholarships, upload documents and track the status of their scholarship application. Role-based dashboards for administrators, faculty and CHED staff also allow the management of applications, review of ranking, monitoring scholars and reports. The Application Layer, which is implemented in Python using FastAPI or Flask, includes business logic for the system. It checks the validity of students' registration and application information with the simulated Registrar Information System, handles the application of scholarships, classifies students, ranks students with the help of scikit-learn, Pandas, NumPy and Joblib using the Random Forest Classifier, and identifies students who do not maintain a 93% retention rate for early warning. The Data Layer, which leverages Supabase and PostgreSQL, serves as a comprehensive persistent storage solution for storing student profiles, scholarship applications, academic records, ranking results, decisions, alerts, and notifications, and enables real-time data sync and role-based access control.

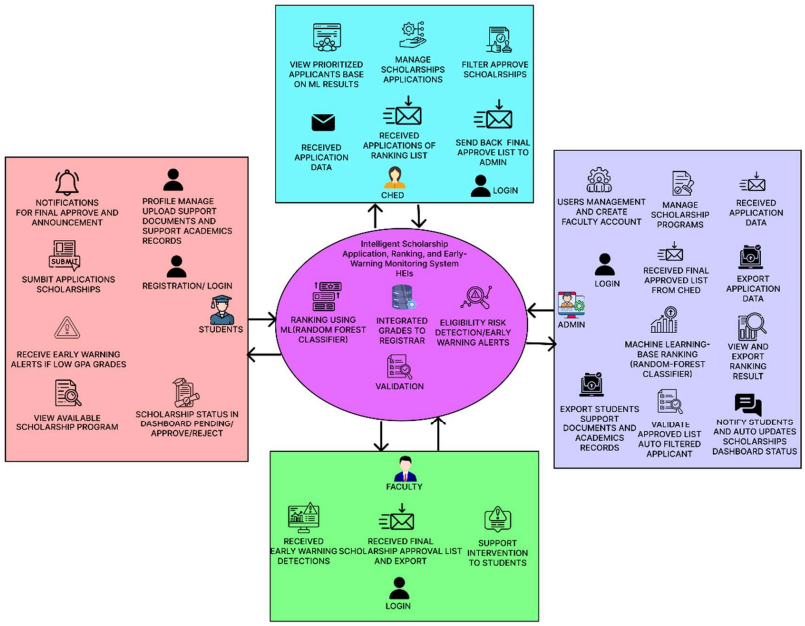


Figure 2. System Architecture Diagram

The system architecture of the proposed platform is depicted in figure 2. The Presentation Layer **(React.js/Vite)** interfaces with the Application Layer **(FastAPI/Flask)** via HTTP/REST API, whereas the Application Layer interfaces with the Data Layer **(Supabase/PostgreSQL)** for data storage and retrieval. The business logic layer processes student input (including data from profiles, applications and supporting documents), saves the resulting data to the database, and generates student outputs, including ranked lists and results of approvals, as well as early warning alerts. The outputs are then shown in the dashboards of the administrators and faculty members and the CHED Review Module checks the ranking list and returns the final approved beneficiaries for release to the students. The three-tier

architecture is fully separated so that the system is easier to develop, maintain and expand and the architecture closely resembles the conceptual architecture.

This three tier architecture applies the conceptual framework of Input-Process-Output (IPO). The Presentation Layer receives user data and provides the needed outputs from the system, the Application Layer process data for the purpose of validation, ranking, approval, and monitoring, and the Data Layer records input and output data during the system's lifetime.

It has four user modules. Using the Student Module students can register, apply for scholarships, administer their profiles, upload documents, monitor application status and get notifications and early-warning alerts. The Administrator Module is used to maintain user accounts, information about the CHED scholarship programs, application records, ranking results, approved lists, and to send the ranking result to the CHED for review. The Faculty Module provides early warning alerts for students at risk of academic failure and provides assistance for interventions. The CHED Review Module processes applications and ranked lists, filters and approves qualified applicants and returns the approved list of applicants to the administrator.

Each user role in these modules has specific functions as described below:

Students

Registration/Login – Establishment of student account and authentication with the mock-up Registrar Information System.

Review Scholarship Programs – explore scholarship applications.

Submit Scholarship Application – Completing and submitting scholarship applications.

Manage Profile and Upload Documents – Uploading supporting documents and academic records to complete student profile.

Track Scholarship Status – Finding out the status of an application, e.g., Pending, Approved or Not Approved.

Receive Early-Warning Alerts – Getting notified when a scholar's GPA falls below the maintaining requirement.

Be notified and be given a Final Approval – Getting notified when a scholarship has been accepted and a Final Approval notification when a scholarship has been awarded.

Administrator

Login – To get into the system with valid credentials.

Manage Users and Faculty Accounts – Managing accounts and creating faculty accounts.

Manage Scholarship Programs – Creating, editing, or updating scholarship programs.

Application Data – Receiving submitted application records for review.

Export Application Data – Exporting application records for reporting or submission.

Execute a Machine Learning-Based Ranking – Producing a ranked list of applicants with Random Forest Classifier.

View and Export Ranking Results – Viewing and exporting the ranking list generated.

Check Supporting Documents and Academic Records – retrieve uploaded documents to verify.

Validate CHED-Approved List – Validation of the final approved applicant list.

Notify Students and Update Scholarship Status – Sending notifications and updating the status of the Scholarship dashboard.

View and Export Support Documents – To verify documents manual in case and have documents to submit to CHED in case emergency can't contact students.

Get Final Approved list from CHED – Final list of approved scholars will be given to the students.

Faculty

Login – Logon to the system, using the username/password provided by the administrator.

Get Early-Warning Alerts – Students reviewed to determine academic concern.

Provide Student Intervention Support – Giving academic counseling and support to at-risk scholars.

View and Export Final Scholarship Approval List – View and Export Approved list of Scholars.

CHED

Login – Logon to the scholarship review portal with assigned credentials.

Get Application Data – Check the completeness of application records.

Get Ranking List of Applicants – Viewing the ranked list of applicants created by the system.

Review ML Based Applicant Prioritization – Examining the randomized forest classed applicants and supporting the decision making process.

Manage Scholarship Applications – Applications being assessed.

Filter & Approve Scholarship Applicants – Identifying and approving qualified applicants.

Return Final Approved List to Administrator – Submitting the approved list to the Administrator.

1.4 Objectives of the Study

This study aims to develop an Intelligent Web Based Scholarship Application, Ranking and Early-Warning Monitoring System for St. Peter's College using Random Forest Classifier for ranking the applicants, early-warning monitoring feature to monitor scholar's academic performance and software quality evaluation of the designed system using ISO/IEC 25010.

The study aims to:

1. An Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System will be developed that will feature:

a. Student Module (React.js Frontend)

i. Registration and user login/logout screen.

ii. Information on the different scholarships available.

iii. Submit online applications for scholarships and uploading ITR or family annual income.

iv. Manage profile upload support documents and academics records.

v. Application tracking of scholarship; real-time tracking.

vi. Viewing of notifications, announcements and early-warning alerts.

b. Admin Module (Admin Interface & Backend Services with FastAPI / Flask)

i. Admin login screen and security.

ii. Management of users and creation of faculty accounts.

iii. Development and configuration of Scholarship programs.

iv. Application, export and management of scholarships.

v. Ranking applicants based on their application using the Random Forest Classifier with a machine learning approach.

vi. Validation of lists approved by CHED with the simulated Registrar Information System.

vii. Notification management, report generation and data exporting tools.

vii. View and export support documents and academics records.

c. Faculty Module

i. Faculty Logon interface & authentication.

ii. Early-warning alerts of scholars at risk are viewed.

iii. Monitoring of scholars' academic performance.

iv. Notifying of intervention and academic support messages.

v. Viewing of the finalized approved scholarship list.

d. CHED Personnel Module

i. Create login screen and authentication for CHED.

ii. The administrator will review scholarship applications sent to the school.

iii. The showing of results for machine learning rankings.

iv. Qualified scholarship beneficiaries to be approved and selected.

v. Providing the final list of scholarships and its return.

e. Core System Functions

i. Data validation and eligibility screening – Student identity, enrollment eligibility, academic records, and scholarship eligibility checked through simulated Registrar Information System.

ii. Machine learning based ranking – applicants' ranking with financial need based on relevant factors using the Random Forest Classifier.

. Early-warning monitoring: Identify students whose academic performance is less than the predetermined 93 percent attendance and creation of alerts for timely intervention.

iv. Secure storage of student profiles, applications, supporting documents, academic records, ranking results, alerts, and more, with role-based access control provided by Supabase database management.

2. Perform Software Quality Assessment of the developed system using ISO/IEC 25010 from the point of view of IT experts, scholarship administrators, faculty and students with regard to:

a. Functional Suitability

b. Reliability

c. Usability

d. Performance Efficiency

e. Security

1.5 Scope and Limitations of the Study

This study focuses on the development, testing, and evaluation of the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System for St. Peter's College in Iligan City, Philippines. The study was conducted during Academic Year 2025–2026 and involved 44 respondents composed of 33 student scholarship applicants, 10 faculty members, and 1 school administrator who participated in the system evaluation.

The study uses the Random Forest Classifier as the only machine learning model for ranking scholarship applicants. The ranking process is based on the applicants' financial information, particularly the annual family income reflected in the Income Tax Return (ITR), and no comparison with other machine learning algorithms was conducted.

The scope of the study includes the development of a web-based scholarship management system with student, administrator, and faculty modules. The system supports student registration and validation through the simulated Registrar Information System, scholarship application submission, document upload, eligibility screening, applicant ranking, scholarship status tracking, notifications, announcements, early-warning monitoring, and report generation. A simulated CHED review and approval process is also included to represent the evaluation of ranked applicants. The software quality evaluation of the developed system has been done on the basis of the characteristics of the ISO/IEC 25010: Software Suitability including functional suitability, reliability, usability, performance efficiency, and security. This study does not address other ISO/IEC 25010 characteristics including Compatibility, Maintainability and Portability.

The limitations of the study are as follows: the system is accessible only through a web browser; it does not include a mobile application; it cannot directly connect to CHED or other external institutional databases; it uses only the Random Forest Classifier and does not compare other machine learning or deep learning models; the CHED review and approval process is simulated within the system and was not tested with actual CHED personnel; the system is limited to St. Peter's College and is not intended for multicampus implementation; and the study does not measure long-term outcomes such as retention, graduation rates, or scholarship performance beyond the system evaluation period. In addition, some scholarship procedures still require face-to-face transactions at the Scholarship Office, including submission of printed valid IDs, physical signatures on scholarship agreements, and the actual release of

scholarship funds. The Announcement feature is used only to inform applicants of schedules and requirements.

1.6 Significance of the Study

The proposed Intelligent Web-Based Scholarship Application, Ranking and Early-Warning Monitoring System will enhance the efficiency, accuracy and transparency of scholarship administration at St. Peter's College. The study will be expected to have an impact on the following stakeholders as it integrates the processing of scholarship applications, validation of applicants, machine learning based ranking, academic monitoring and notification services into a single platform:

Students

Students will enjoy much easier access to information on scholarships, the ability to apply online, get scholarship status updates in real-time, and receive automatic emails whenever there are scholarship requirements or application results. Finally, the Early-Warning Monitoring will alert the student to academic problems that could impact whether or not they are eligible for a scholarship and can take appropriate action in a timely fashion to turn things around and enhance their chances for success.

Scholarship Administrators

Centralized applicant records, automated validation, and organized scholarship information will streamline and improve the efficiency of scholarship processing. The use of the Random Forest Classifier for applicant ranking will also accelerate the evaluation of applications, provide more consistent and objective evaluation, and reduce the manual workload.

Faculty Members

Early-Warning Monitoring feature will help faculty to identify students who are at risk of losing a scholarship. This will allow faculty to offer timely academic guidance, counseling and intervention to help students fulfill scholarship requirements and enhance their academic abilities.

St. Peter's College

The institution will enjoy better data management, workflow automation, and centralized data record keeping. The system can facilitate more effective communication between stakeholders, increase transparency in scholarship evaluation processes, and give stakeholders reliable information for monitoring the scholarship recipients and support institutional decision making.

The Commission on Higher Education (CHED) Personnel

CHED staff will have access to organized scholarship records, applicant information and ranking results. The machine learning ranking process can assist in the more systematic and objective evaluation of the scholarship applicants, according to the criteria of eligibility and their financial needs, that can help to enhance the scholarship evaluation process.

Future Researchers

The study can serve as a reference for scholars who wish to develop a scholarship management system, a system of academic early-warning monitoring, as well as machine learning applications for educational decision-making in the future. This study's results can offer useful information and be a foundation for continuing research for better managing scholarships and educational information systems.

1.7 Definition of Terms

The following definitions are provided for clarity of concepts used when developing the Intelligent Web Based Scholarship Application, Ranking and Early-Warning Monitoring System. Each term is discussed in its general form and, when appropriate, in the context of its use in this study. The terms are arranged alphabetically, beginning with Alumni and ending with User.

Alumni — A person who has completed a period of study and progressed to a higher year level within an institution (Merriam-Webster, n.d.). In this study, Alumni (or Continuing Student) is an applicant who is renewing a scholarship, based on College GPA.

Applicant — An individual who submits a formal request to be considered for a program or benefit (Merriam-Webster, n.d.). In this study, Applicant: Student applying for a scholarship using the proposed system.

Commission on Higher Education (CHED) — The government agency created under Republic Act No. 7722 (1994), otherwise known as the Higher Education Act of 1994, mandated to formulate policies, set standards, and oversee both public and private higher education institutions in the Philippines. In this study, The members of a CHED are those who are responsible for reviewing the applications for scholarships, assessing the results of the rankings, approving the beneficiaries of the scholarships, and submitting the final rank list to the administrator.

Early Warning Alert Detection (EWA) — A monitoring process used to identify individuals who may be at risk based on predetermined criteria (Hu et al., 2014). In this study, EWA is a feature of the system designed to keep track of the scholars'

academic performance and alert the scholars if their grade is under 93% of maintaining grade.

Eligibility Screening — The process of determining whether an applicant meets the requirements to qualify for a program or benefit (Al-Ayyubi & Maulana, 2023). In this study, Eligibility Screening is the process of confirming a student's identity, enrollment, academic and financial information to meet the requirements of a scholarship.

Faculty Member — An employee who is part of the teaching staff of an educational institution (Merriam-Webster, n.d.). In this study, Faculty Member the user who is notified of alerts of at-risk scholars, who keeps track of at-risk scholars, and who offers academic support or intervention.

Freshman — A student in the first year of study at the college level (Merriam-Webster, n.d.). In this study, Freshman is an applicant category that is determined by the applicant's Senior High School General Weighted Average (GWA).

Higher Education Institution (HEI) — A post-secondary or tertiary institution offering academic degree programs, as recognized and regulated under Republic Act No. 7722 (1994). In this study, HEI stands for the St. Peter's College wherein the proposed system will be implemented and evaluated.

ISO/IEC 25010 — An international standard for the quality of software products, which includes eight Quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (ISO/IEC 25010:2011). In this study, The framework to assess the system developed is known as ISO/IEC 25010 which is related to functional suitability, reliability, usability, performance efficiency, and security.

Machine Learning — A branch of artificial intelligence in which systems learn patterns from data to make predictions or decisions without explicit programming (Mitchell, 1997). In this study, The process of ranking the candidates for the scholarships based on the annual family income is called Machine Learning.

Random Forest Classifier — An ensemble machine learning algorithm that builds multiple decision trees and aggregates their outputs to produce more accurate and stable classifications (Buslim et al., 2023). In this study, Random Forest Classifier is the algorithm used to rank the applicants for scholarships according to their Income Tax Return (ITR), annual income of parents and other pertinent financial and academic details.

Ranking System — A method of ordering applicants or records according to specific criteria or evaluation factors (Buslim et al., 2023). In this study, The process of arranging the applicants of a scholarship based on their financial need and other factors from the lowest to highest is called Ranking System and it is done by the Random Forest Classifier.

Reliability — The capability of a software product to maintain a specified level of performance under specified conditions (ISO/IEC 25010:2011). In this study, Reliability means consistency and stability of the proposed system's performance.

Scholar — An individual who receives financial support through a scholarship program (Merriam-Webster, n.d.). In this study, Scholar is any student who receives a scholarship and is under the control of the suggested monitoring system.

Scholarship Administrator — The individual responsible for managing scholarship-related activities and records (Remegio & Casquejo, 2025). In this study, Scholarship

Administrator the user who manages scholarship programs, reviews, validates, ranks and informs applicants and scholars.

Scholarship Application — A formal request submitted by an individual to be considered for a scholarship program (Al-Ayyubi & Maulana, 2023). In this study, Scholarship Application the information and paperwork submitted in the system for scholarship consideration.

Usability — The degree to which a system can be used by specified users to achieve goals effectively, efficiently, and with satisfaction (ISO/IEC 25010:2011). In this study, Usability is the functionality and friendliness of the offered system from users' point of view.

User — A person who interacts with or accesses a system (ISO/IEC 25010:2011). In this study, User is the students, administrators, faculty members and CHED personnel who use the proposed system.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

The review of related literature and studies that are relevant to the development of the Intelligent Web Based Scholarship Application, Ranking, and Early-Warning Monitoring System for St. Peter's College in Iligan City, Philippines are presented. It has four main themes: Web-Based Scholarship Management Systems, Early-Warning Systems and Academic Risk Prediction, Integration of Scholarship Management, Software Quality Standards Using ISO/IEC 25010, and Machine Learning Ranking, and Early-Warning Monitoring Systems. Themes give the conceptual and practical context to the study in terms of digital scholarship administration, ranking of applicants, academic monitoring, and software quality assessment.

2.1 Web-Based Scholarship Management Systems

Web-based scholarship management system is an information system, which is helpful to digitize and enhance the application and validation process, track and record of scholarships. These systems can mitigate manual paperwork, enhance access to scholarship services and facilitate effective scholarship administration with centralized online systems.

studies indicate that web-based scholarship systems enhance accessibility and administrative efficiency by giving students a simple way to submit applications online, and providing the administration with an easy way to handle records electronically. The study by Al-Ayyubi and Maulana (2023) revealed that web-based

platforms for scholarships enhance data management in application processing and increase accessibility for scholarship services. Likewise, Ablen et al., (2025) concluded that the SMART-Scholar system minimized administration burden and transparency by automating applications. Berido et al. (2023) have also shown that machine learning with statistical methods can enhance the systematic assessment of the eligibility of applicants. While Al-Ayyubi and Maulana (2023) and Ablen et al. (2025) concentrate primarily on improving the application submission process and ensuring administrative transparency, Berido et al. (2023) further automates the process of assessing eligibility itself, without moving on to ranking applicants or tracking their status once they are approved.

of the existing scholarship systems however, are limited to the online application, record keeping and administrative monitoring. They typically do not involve an intelligent ranking of applicants, or on-going monitoring of approved scholars. This study aims to fill that gap by proposing a Scholarship Application Management, Applicant Validation, Machine Learning based Ranking and Early-Warning Monitoring System for all the above applications in one platform.

2.2 Early-Warning Systems and Academic Risk Prediction

Early-warning systems are information systems that, by using data including student grades, attendance, and other indicators are designed to identify students who could be at risk of academic difficulty. These systems support institutions to deliver timely interventions before students get into serious academic issues.

Recent research demonstrates that early-warning systems can be valuable tools for identifying students at risk and academic intervention. The study by Hu et al. (2014) and Qu et al. (2022) highlighted that the analysis of students' performance and

behavioural records should be able to reveal patterns that help forecast future academic difficulty. Plak et al. (2022) identified that EWS can support institutions to identify students who require support interventions; Boudjehem and Lafifi (2024); and Bañeres et al. (2023) established that ongoing monitoring combined with timely intervention can decrease drop-out rates and enhance the effectiveness of student support.

Even with these advantages, the majority of early-warning research is on academic monitoring instead of scholarship-specific issues. They are not often concerned about scholarship qualification, keeping good grades and scholarship tracking. However, the call to support selection and continuous monitoring of vulnerable individuals was made explicit by Liz-Domínguez et al. (2019), and has been underdeveloped in the literature available on scholarships. To fill that void this study has integrated early-warning monitoring into the scholarship administration system to monitor scholar's progress and to send alerts when it is not being met if scholars are not meeting grade standards.

2.3 Integration of Scholarship Management, Machine Learning Ranking, and Early-Warning Monitoring Systems

Evidence has shown Scholarship administration tools, machine learning applicant ranking models, and early-warning monitoring tools are designed to enhance decision-making and student outcomes, but due to these technologies being created in silos, application administration, ranking and scholarship management don’t “speak” well.

In addition, decisions support systems help the administration of scholarship by systematically managing the applicant data and facilitating the selection process

through proper analysis as reported by Remegio and Casquejo (2025). Furthermore, by using various machine learning methods like classification and prediction of applicants and for scholarships selections by Chaudhari et al. (2022), Paramesha et al. (2024) and Kumar and Vaishnavi (2025) also confirmed the same. Similarly, from a selection of applicants using machine learning, applications are evaluated more systematically as reported by Fajardo et al. (2024) and Kanwal et al. (2024).

However, the majority of existing works focuses on one part of scholarship administration like the system that handles application, scholarship applicant ranking or monitoring scholar. It seldom comprises all 3 applications of scholarship application processing, candidate rank, and scholar early-warning monitoring collectively to one scholar administration. This work could fill the gap since it integrates web-based scholarship administration, Random Forest-based scholarship candidate rank and scholar early-warning monitoring into one portal.

Random Forest Classifier for Application Ranking In this study, the application of the Random Forest Classifier was considered to rank the applicants since this machine learning model can observe several variable factors and classify the decision based on learning a trend in available data. With reference to this, Random Forest Classifier was employed to rank scholarship applicants with financial needs particularly the ITR information and other variables. Buslim et al. (2023) state that scholarship selection can attain objectiveness and a more consistent decision through an ensemble classifier system. Hence machine learning can play a significant role in application ranking.

2.4 Software Quality Standards Using ISO/IEC 25010

An adequate Software Quality Evaluation can form part of the developing phase of a system. Evaluation is necessary to verify that the system can effectively fulfill its expected functionalities and that it can reliably support its users. ISO/IEC 25010 is commonly referred to by practitioners because the characteristics used are clearly identifiable.

The ISO/IEC 25010 model includes eight major quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. These characteristics provide a structured basis for evaluating whether a system meets user needs and performs well in practice.

Several studies have used ISO/IEC 25010 to evaluate information systems and web-based applications. The standard is especially useful in educational systems because it helps assess functionality, usability, efficiency, reliability, and security. In this study, the proposed system will be evaluated using selected ISO/IEC 25010 characteristics, namely functional suitability, reliability, usability, performance efficiency, and security.

2.5 Summary

The surveyed literature reveals that web-based scholarship management systems promote greater access to scholarship services and administrative effectiveness, but generally fail to address the administrative needs for record management in addition to the processing of scholarship applications. The literature concerning academic early warning systems reveals that monitoring student achievement can help locate those students who are at risk but fails to address scholarly concerns nor scholarship administration issues, and the scholarship

management process and the machine learning approaches found indicate that machine learning algorithm such as Random Forest can potentially help evaluate the ranking of applicant; however this is rarely explored in a joint manner for both scholarship management and scholar monitoring together.

The studies cited are showing the relevance of using ISO/IEC 25010 for evaluating the quality attributes of a software namely functionability, reliability, usability, performance efficiency, security and so on. In no cases of the reviewed study the web-based scholarships management, Random Forest and ranking approaches combined with the approach of early-warning monitoring were combined into a system for St Peter's College; thus provides a rational basis for creating of the Intelligent Web-Based Scholarship Application, Ranking and Early-Warning Monitoring System.

CHAPTER 3

RESEARCH METHODOLOGY

In this chapter, the research methodology applied in designing, developing and evaluating the Intelligent Web-Based Scholarship Application, Ranking and Early-Warning Monitoring System for St. Peter's College is presented. It has explained the research design, the strategy applied in the development of the system, the development phase, the modeling of the system and the design, the setting of the research, the research respondents, the research instrument, the data collecting procedure, the scoring procedure of the system in evaluating test, the treatments applied and ethics. It also detailed the evaluation of the developed system based on some traits in ISO/IEC 25010 Software Quality model.

3.1 Research Design

Intelligent web-based scholarship application, ranking and early -warning monitoring systems for St. Peter's College was conceived, developed and tested with a development research design, in which according to Richey and Klein (2007), development research encompasses the design, development, and testing of the technological products in an organized systematic manner with the view of ensuring its effectiveness and consistency. This design fits the current investigation because it is the plan of how this system can be designed and, then tested for the attainment of the goal outlined in Section 1.4 of this research paper.

By utilizing the developmental research design, the developers succeeded in developing a ready-to-use system, assessing it with the identified quantitative quality criteria, and including the final end-user's comments in their modification processes. They have, by that way successfully addressed the scholarship management concerns of the institution most especially on academic early-warning monitoring, ML-based scholarship rankings, and validation of scholarship applicants. ISO/IEC 25010 Software Quality Model has been employed by determining the quality level of each determined criteria as an ISO 25010 software characteristic namely: functional suitability, reliability, usability, performance efficiency, and security in which a quality standards set were applied.

3.1.1 SDLC Model

In developing the Intelligent Web-Based Scholarship Application, Ranking and Early-Warning Monitoring System, Iterative Software Development Life Cycle (SDLC) Model was adopted as the system development life cycle framework. An Iterative SDLC Model is divided into the processes: planning, analysis, design, implementation, testing, and evaluation, and were chosen because it allows the system to be refined and improved via continuous development cycle and user feedback.

The iterative methodology fits well with the presented system as it comprises of multiple linked modules like administration of database, analysis of student academic performance, handling of scholarship application, applicant ranking and management of notification to the users. It provides ability to make amendment to the functional of the system and analyse performance and integrate with the result of system iteration with feedback. Moreover Random Forest Classifier that is employed to rank the scholarship applicants on basis of financial criterion can also be incrementally used and refined.

Iterative Approach of Development Methodology Such multiple modules will be connected to each other i.e. Database administration module, students' academic performance report section module, scholarship application section module, applicant ranking and management section module, as well as the user notification module in the project and this approach allows re-arrangements in module functionality and development of modules to assess current performance by evaluating the output of each system version against its user feedback to re-start another round, where the machine based classifier Random Forest Classifier adopted to categorize students based on their financial needs for their eligibility may also developed and enhanced iteratively.

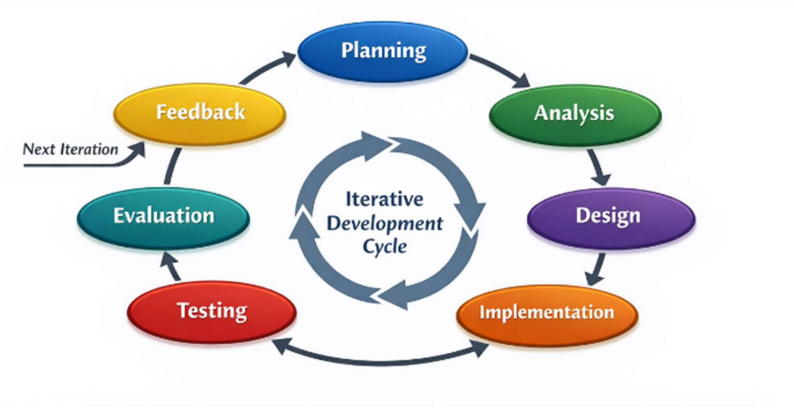


Figure 3. Iterative SDLC Model for the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System

This figure 3 shows SLDC model the process of designing and developing the Intelligent Web-Based Scholarship Application Ranking and Early Warning Monitoring System, the iterations as mentioned above were adhered to. The Iterative Software Development Life Cycle (SDLC) Model involves six different iterating processes planning, analysis, design, implementation, testing and evaluation.

Planning Phase

The current scholarship management process in St. Peter's College has been identified and analyzed to find operational problems and discrepancies due to the present manual processing of scholarship application and absence of proper scholar monitoring system in the planning phase. From the identified problems, project objectives, its boundary, functional requirements and technical requirements were fixed. The various stakeholders such as students, teachers and administrators requirements were considered and identified.

Analysis Phase

The analysis phase was concerned with identifying and specifying the functional and non-functional requirements for the new system. The workflow within the existing scholarship application process was investigated in detail in order to find workflow bottlenecks and scope for enhancement. Key information, such as student personal data, registration status, grade Point Average, fee account and various application related documents were analyzed to produce the system requirements specifications, as well as the validation criteria/rules which govern that only suitable applicants shall have and shall be able to utilize the system.

Design Phase

During the design stage, the entire system architecture was structured based on a three-tier architecture, comprising of the Presentation Layer, Application Layer, and Data Layer. The database structure, user interface, and process flows were conceived to enable effective scholarship application, scholarship applicant ranking, and academic monitoring. Models such as the Input-Process-Output (IPO) Model, Use Case Diagram, Context Diagram, Data Flow Diagram (DFD) and Entity Relationship Diagram (ERD) were created to better show system process flow, data flow and

relationships in the database, respectively. The Random Forest Classifier was also included in the design, for the intelligent ranking of scholarship applicants according to their financial capabilities. All components of the UI were developed with student, staff, scholarship administrator in mind to ensure that it's responsive, accessible and user-friendly.

Implementation Phase

During the implementation phase, all the specified system modules were built and integrated to create an entire web application. The backend services are implemented in Python using FastAPI or Flask framework. The frontend is created with React.js using Vite framework. The cloud-based database management system Supabase (based on PostgreSQL) was used to manage and storing the application data. The Random Forest Classifier was implemented using scikit-learn ML library where Pandas, NumPy and Joblib were used for data pre-processing and deploying the model. The modules specified in section 1.4 were combined to provide the functioning system where the module includes user management, application handling, applicant sorting, early warning monitoring system, notification and report generation.

Testing Phase

The developed system was verified to check if both functional and nonfunctional requirements had been achieved. The functional and technical defects recorded during the test processes were rectified to ensure a robust and efficient functioning of the system. Various activities including checking for functional readiness, stability/robustness of the system, user friendly environment, efficiency in performance and security of the system were completed prior to release the system.

Evaluation Phase

The system developed in the solution phase was tested during evaluation process by some selected qualities according to ISO/IEC 25010 Software Quality Model framework by respondents that rated the system according to functional suitability, reliability, usability, performance efficiency and security. The result acquired by this evaluation determines whether the proposed developed system has achieved the quality level required or if it has met the non-functional and functional qualities of the identified scholarship management system.

3.1.2 Physical Environment and Resources

For the development and implementation of Intelligent Web-Based Scholarship Application, Ranking and Early-Warning Monitoring System for St. Peter's College; sufficient resources were required so that the software development, testing, deployment and evaluation could be carried out efficiently. The resources available supported software development, testing, deployment and evaluation by providing the required computation environment and development tools. The requirements for hardware, software and technologies use in the research are presented in the following sub-sections.

Hardware Requirements

The hardware requirements to implement, test, and deploy system. Provided adequate resources to the required performance, memory size, storage, and network connections needed to develop software, store databases, to conduct machine learning and to test the system as described in Table 1.

Table 1. Hardware Requirements

Table 1 shows the hardware use to develop of Intelligent Web-Based Scholarship Application, Ranking and Early-Warning Monitoring System.

Component	Specification	Purpose / Justification
Processor (CPU)	Intel Core i5 (8th Generation) or higher	Supports system development, backend processing, and machine learning computations.
Memory (RAM)	8 GB or higher	Ensures smooth execution of development tools, browsers, and local servers.
Storage	256 GB SSD or higher	Provides fast and reliable storage for system files, databases, and development resources.
Monitor	1080p resolution or higher	Allows clear visualization of system interfaces and the development environment.
Internet Connection	Stable broadband connection	Required for cloud database integration, API communication, and real-time system operations.

Software Requirements

The software components that are required in developing, operating, maintenance, and testing of the proposed system are operating system, developing tools, database management system, and other software that are required to build web based scholarship application and ranking & early-warning monitoring system. Table 2 lists the software requirements.

Table 2. Software Requirements

Table 2 shows the hardware use to develop of Intelligent Web-Based Scholarship Application, Ranking and Early-Warning Monitoring System.

Category	Tool / Technology	Purpose / Role
Operating System	Windows 11 (64-bit)	Provides a stable environment for development tools and system operations.
Frontend Framework	React.js with Vite	Used for building responsive and dynamic user interfaces.
Styling Library	Tailwind CSS	Provides efficient and consistent user interface styling.
Markup Language	HTML	Structures web content and interface elements.
Backend Framework	Python (FastAPI or Flask)	Handles system logic, authentication, validation, and data processing.
Machine Learning Library	Scikit-learn	Implements the Random Forest Classifier for applicant ranking.
Data Processing Libraries	Pandas and NumPy	Used for data preprocessing and numerical computations.
Model Deployment Tool	Joblib	Saves and loads trained machine learning models.
Database	Supabase (PostgreSQL)	Stores and manages system data securely and in real time.
Version Control	Git and GitHub	Supports code versioning, collaboration, and project management.
Web Browser	Google Chrome and Mozilla Firefox	Used for system testing and interface evaluation.

3.1.3 Tools and Techniques Used in the Study

An architectural representation of the proposed system was modeled using structured modeling techniques to explain the system structure, functions, data, and relationship of the system database. The above models are developed as a reference for the system design, development and integration. The diagramming techniques used for modeling the system were: Use Case Diagram, Context Diagram, DFD, and ERD.

Use Case Diagram

This section illustrates how the users interacts with the core features of the planned software systems. It guides to select the role members of the planned system and their relevant tasks. The use case diagram describes the features offered by the software system.



Figure 4. Use Case Diagram

The four actors of the system are as follows: Administrator, Student, faculty, and CHED (as shown in figure 4). There are several activities in which the role of actors is crucial in the process of scholarship application processing and application evaluation notification.In the process of processing and evaluating the applications for scholarships and notification, there are several activities that the role of actors is crucial. The diagram shows the interactions between these four actors and the scholarship management system, at the same time, each actor has different

functionalities that they use for the different aspects of the job and the responsibilities they have. The role of the Administrator actor is mainly to oversee the system, manage their accounts and implement the machine learning ranking process. The Student actor is the main user who registers, applies for and tracks their scholarship. The Faculty actor is to facilitate academic monitoring and intervention for at risk scholars, and the CHED actor is to act as the external government agency in charge of final evaluation, approval and release of the list of scholars. All these actors are part of a coordinated flow that guarantees transparency, accuracy, and accountability for the scholarship application and evaluation process, from student registration to final approval, monitoring, and more.

Table 3. Administrator Use Cases

The following table details the administrator use case name, actors and description from UC01-UC15. It shows the administrative tasks, such as account and role management, setup of the scholarship program, configuration of the deadlines, document verification, and execution of the Random Forest Classifier to produce a ranking of the applicants based on financial need.

Use Case ID	Use Case Name	Actor	Description
UC01	Login	Administrator	Access the scholarship management system securely.
UC02	Manage Users	Administrator	Manage student and faculty accounts within the system.

Use Case ID	Use Case Name	Actor	Description
UC03	Create Faculty Account	Administrator	Create faculty accounts for monitoring and intervention purposes.
UC04	Assign Roles	Administrator	Assign user permissions and responsibilities.
UC05	Remove Duplicate/Inactive Accounts	Administrator	Detect and deactivate duplicate or inactive accounts.
UC06	Export Application Data	Administrator	Export application records for manual checking and reporting.
UC07	Receive Application Form Data	Administrator	Receive and review scholarship applications submitted by students.
UC08	Manage Scholarship Programs	Administrator	Manage scholarship programs and application forms.
UC09	Create Scholarship Programs	Administrator	Create scholarship opportunities available to students.
UC10	Set Deadlines	Administrator	Set scholarship application deadlines.

Use Case ID	Use Case Name	Actor	Description
UC11	Run Machine Learning Ranking	Administrator	Execute the Random Forest Classifier to generate applicant rankings based on financial need, using the annual family income submitted through the ITR.
UC12	View and Export Rank List	Administrator	View and export applicant ranking results generated by the machine learning model.
UC13	View Support Documents	Administrator	View uploaded supporting documents (COR, Valid ID, Signature Form) submitted by students for verification purposes.
UC14	Export Support Documents	Administrator	Export/download uploaded supporting documents for physical submission to CHED.
UC15	View Support Academic Records	Administrator	View uploaded academic records (SHS GWA or College GPA) submitted by students as supporting proof.

Table 4. Student Use Cases

The following table lists Student use case name, actors and description for UC01 – UC11. It shows the student's process of signing up for an account, validating it, browsing for scholarships, submitting applications, uploading supporting documents and tracking the application status on the dashboard.

Use Case ID	Use Case Name	Actor	Description
UC01	Register Account	Student	Create an account for scholarship application.
UC02	Validate	Student	Validate account information using the simulated Registrar Information System database.
UC03	Select Applicant Type	Student	Select applicant category such as Freshman or Alumni.
UC04	Freshman or Alumni	Student	Specify the applicant type during registration.
UC05	Login	Student	Access the scholarship application system securely.
UC06	Manage Profile	Student	Update and maintain personal account information and profile details.

Use Case ID	Use Case Name	Actor	Description
UC07	View Available Scholarship Applications	Student	Browse available scholarship programs and opportunities.
UC08	Submit/Apply Scholarships Applications	Student	Complete and submit scholarship application forms.
UC09	Upload Support Documents	Student	Upload required supporting documents (COR, Valid ID, Signature Form) for verification purposes.
UC10	Upload Support Academic Records	Student	Upload academic records (SHS GWA or College GPA) as supporting proof.
UC11	View Application Status in Dashboard	Student	Monitor scholarship application status such as Pending, Approved, or Not Approved.

Table 5. CHED Use Cases

The following table displays the use case name, actors and description of use cases UC01-UC06. It shows the role of CHED as the approving authority, accepting

application data, checking on the list of scholars prioritized by the ML, filtering and approving the scholars and passing the list of approved scholars to the institution.

Use Case ID	Use Case Name	Actor	Description
UC01	Login	CHED	Access the scholarship management portal securely.
UC02	Receive Application Data	CHED	Receive scholarship application records from the institution.
UC03	View Prioritized Applicant Base on ML Ranking Result	CHED	Review applicants prioritized through machine learning ranking results.
UC04	Scholarships Management	CHED	Manage scholarship applications and approval processes.
UC05	Filtered/Approve Scholarships	CHED	Evaluate, filter, and approve scholarship applicants.
UC06	Send Back Final Approve List to HEIs	CHED	Return the final approved scholarship list to the institution.

Table 6. Faculty Use Cases

This is the Faculty use case name, actors and description from UC01-UC04. It shows Faculty monitoring, which includes receipt of the final approved list of

scholarships, provision of academic intervention, and the receipt of early warning alerts regarding students who are in danger of losing their scholarships.

Use Case ID	Use Case Name	Actor	Description
UC01	Login	Faculty	Access the monitoring and intervention system securely.
UC02	Receive and Export Final Approved Scholarship List	Faculty	Receive and export the final approved scholarship recipients list.
UC03	Support Interventions to Students	Faculty	Provide academic monitoring and intervention to students at risk of losing scholarship eligibility.
UC04	Receive Early-Warning Alerts	Faculty	Receive alerts regarding students whose academic average falls below the required maintaining grade.

Context Diagram

The context diagram depicts the system as a single process and how it exchanges and interacts with the Student, Faculty, Administrator, and CHED entities. It provides a view of the information that each actor shares with the system (e.g., applications, user management requests and decisions) and also of the information that the system returns to the actors (e.g., validated applications, ranking results,

notifications, early-warning alerts), ensuring that all the stakeholders are kept informed during the scholarship process.

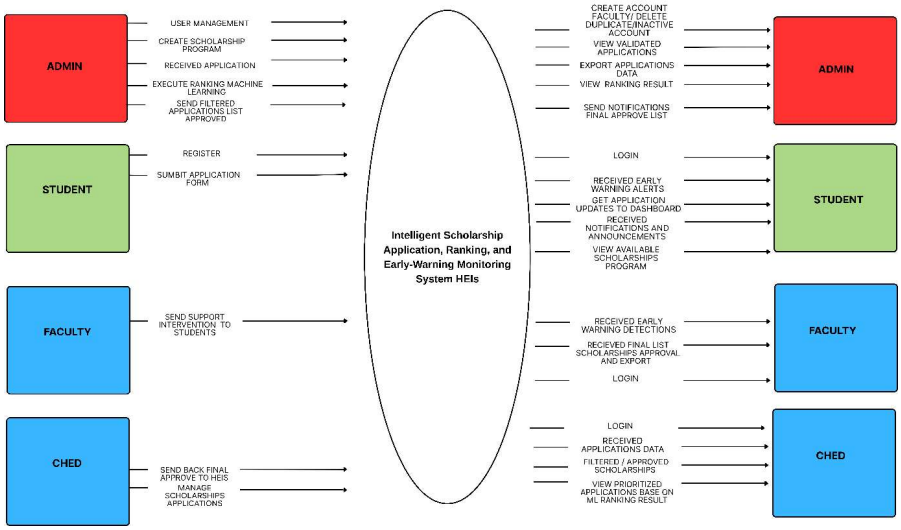


Figure 5. Context Diagram

This figure 5 depicts the context diagram of the system as one process which communicates and interacts with external entities of the system - Administrator, Student, Faculty and CHED. It depicts the inputs and outputs of the system that are contributed by each actor, such as for instance a ranking result, a notification as well as an early warning alert, but also a scholarship application and a user management request. This diagram depicts the data flow between the Intelligent Scholarship Application, Ranking, and Early-Warning Monitoring System and its four external entities, and the data each entity puts into the system and the data each entity receives in return. On the left side, the Administrator prepares input information, including user management, creating the scholarship program, and issuing the result of the ranking machine learning process; the Student enters the information of registration, application forms, and the Faculty sends the information for intervention support; finally, CHED provides information on the results of the Intervention Support Program and the instructions for the management of the application forms. The system circulates the outputs of the actors on the right side back to the actors, with validated applications, ranking results, notifications, dashboard updates, and early-warning

alerts -- showing two-way flow and keeping every actor informed and synchronized throughout the scholarship process.

Data Flow Diagram

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Data Flow Diagram (DFD) describes how data will flow and be processed inside the system. Data flow diagram also allows users to peek at the data processes and data flow between user, process and data store in the system.

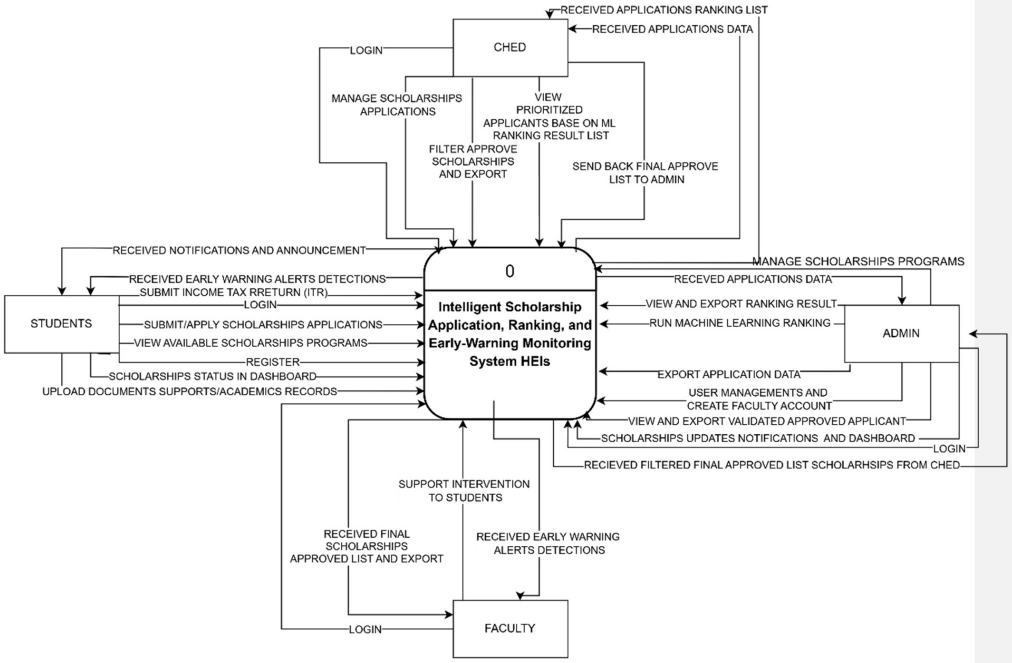


Figure 6. Data Flow Diagram (Level 0)

This figure 6 shows the following is Level 0 DFD of Intelligent Scholarship Application, Ranking and Early-Warning Monitoring System. The DFD shows the process which occurs in the system from the highest and most abstract perspective. The diagram show the inputs and outputs that flows between the system and its external environment.

The user logs in or registers an account. Upon registration, the student data will be validated against the simulated Registrar Information System Database that contains all student's enrollment records and transcripts. The student's personal profile

is then complete with an accomplished of supporting documents, i.e., (Certificate of Registration (COR), Valid ID, and Signature Form), accompanied by its supporting academic records, i.e., (Senior High School GWA - for freshman applicants or College GPA for ongoing/alumni applicants), proof of Academic Performance, respectively. The supporting documents, curriculum, records that are filed under each student account serve as reference files which may be view or downloaded by the administrator such as in submission to the office physically to CHED. Once the student account is complete, student can availing for scholarship, able to apply for the scholarship (Where the scholarship application itself asking him/her to upload IncomeTax Return(ITR) of theirs, and entry of annual income of student wherein its output going to pass through the Random Forest Classifier to be ordered according to how much financial the family needs), check current status, get early warning notices (early warning is to be released if a student's academic grade is less than 93%).

The administrator logs in to the application. Some tasks to be executed by the administrator are user administration, administration of the faculty accounts, Scholarship programs administration, Scholarship application forms administration. The administrator will able to retrieve information of the applications that have been submitted by the students. Administrator will view, download and export students' submitted supporting documents and supplementary student records for verification and purposes and also downloads and exports application details if needed and uses Random Forest Classifier machine learning algorithm to produce ranked students from financial needs obtained from the submitted annual family income. Administrator will also be able to view and download ranked students as well as verify scholars through the registrar database and receives verified scholarly list from CHED for the purpose of notifying the scholars. This system also allows faculty access where they can receive proactive alerts sent through the monitoring module. Faculty alerts to notify which students' average academic standing is below the minimum standard in order to be at risk academically. With this access faculty can intervene, view where students are falling, and which ones are approved scholars. User logs in the system (CHED) manages the Scholarship application process CHED data inputs (student info & ML ranked) and then prioritzies the eligible students to be a scholar and send the selected scholars' list to the administrator. Additionally, it is linked to a test Registrar Information System. Using the connection provided by it, the system will be able to provide:

automatic verification of student; automatic validation of enrollment; check of the grades; determine if the scholars are eligible. Also, through Random Forest Classifier the system can ranks scholars according to their financial necessity and eligibility.

Typically, the system enhances the overall scholarship management process as it provides a user interface where application processing, profile and document management, scoring of candidates, qualification checking, monitoring of the approval of applications, automatic alert and communication between students, administration, teaching staff and CHED is performed thereby solving most of the scholarship management processes being tackled with manual and data tracking systems (and other issues among scholarships).

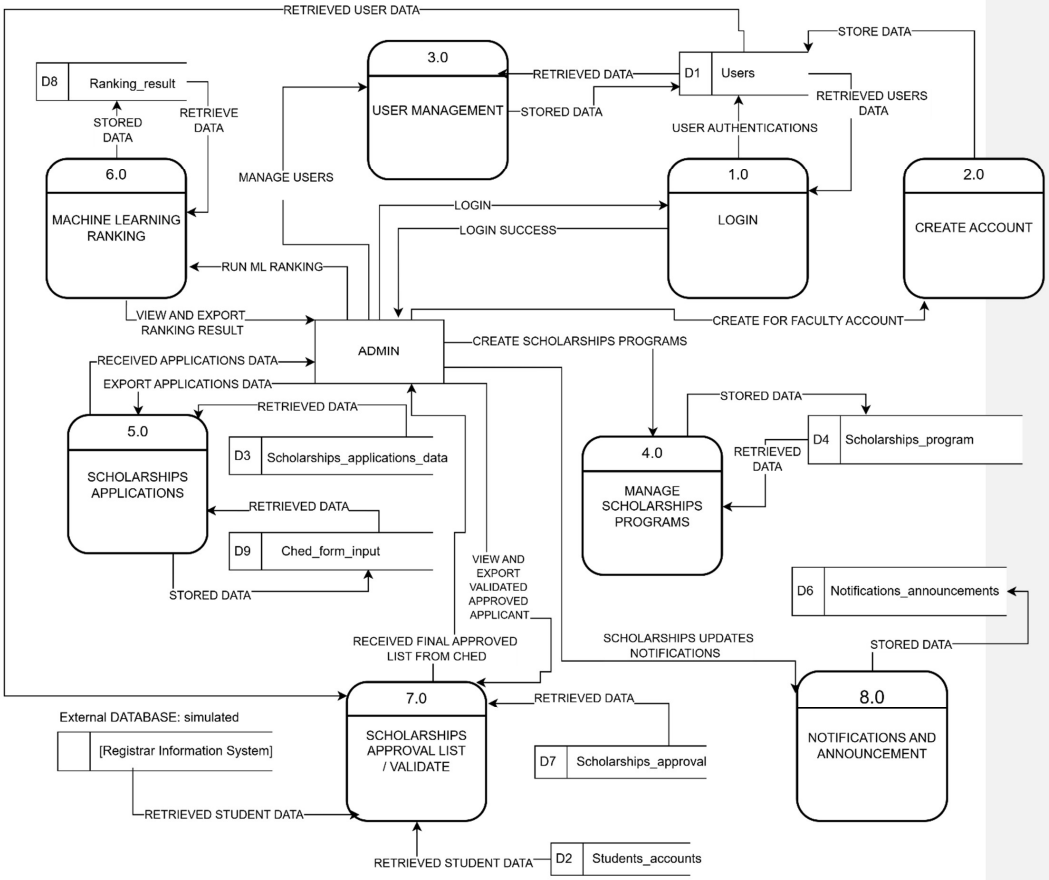


Figure 7. Administrator Data Flow Diagram

Figure 7 shows the DFD of Administrator of Intelligent Scholarship Application, Ranking and Early Warning monitoring System. It covers the functioning where administrator manages User Account Management, Scholarship Program Management, Scholarship Application Management, ML Based Application Ranking, Scholarship Auditing, Notification to Student; also the data transfer between Administrator, the mockups registrar Information System, and internal data stores. The next phase is 1.0 Login in which the administrator logs in to the system using the required username and password. System reads the credentials and authenticates administrator's credentials. If administrator is authenticated, he is allowed to access the system. After the login, the administrator proceeds to 2.0 CreateAccount where the faculty and guidance personnel accounts will be created. The new users are created by the administrator by providing the requested details and are saved in the Users data store to enable further authentication and usage. From here the administrator may proceed to 3.0 User Management, which refers to the management of the user's accounts. This also involves retrieve the users' information from the Users data store in order to enable the administrator modify account information, remove duplicate entries, delete inactive and remove student accounts of those students who were not able to enroll or graduated. The Modified account information are written in the Users data store. Secondly the administrator uses 4.0 Manage Scholarship Programs which provides functionality to create a scholarship program, edit a scholarship program, or delete a scholarship program. The scholarship program can be a new, existing, and removed scholarship with a scholarship description, eligibility requirements and guidelines, due date and application details and so on. • The

Scholarship_program entity will store all the information about the scholarship program. Review scholarship application process administrator. The reviewer checks out the following tasks: 5.0 Scholarship Applications This is the tasks of viewing scholarship applications of selected students. It uses both the Scholarships applications data and the CHED_form_input data stores to access for the information about applications and updates some part that is needed for updating and exporting it for assessment and evaluation. Once a scholarship applications are analyzed the administrator executes 6.0 Machine Learning Ranking. That is where the Random Forest Classifier is used to predict scholarship applicants to a ranking of scholarship candidates according to set criteria for scholarship selection, in this specific regard, the family income input on submitted applications which helps classify the level of financial need. This ranking is then stored in the Ranking_result data store. From 7.0 Scholarship Approval List/Validate, the administrator proceeds to this part once the final list of scholarship students approved by the CHED is available. These approved records are queried from the Scholarships_approval data store, to validate data administrator looks for matches on the students' information from simulated Registrar Information System, Users, and Student_accounts data stores using the Student ID and Full Name, who then become certified as scholarship students only if there is matched students in all records, which they are present students from St. Peter's College. The administrator performs 8.0 Notifications and announcements. The administrator makes notifications in the form of scholarship announces. It notifies the approved candidates for their approved scholarships, updates scholarships, prompts the students to provide mandatory documents and provides intervention to the students. All the notifications are stored in the Notifications_announcements and provided to the students via their dashboards. Finally, the administrator is able to do

9.0 View/Export Support Records. The administrator can access the student's uploaded supporting documents (COR, Valid ID, Signature Form) and the student's supporting academic records (SHS GWA or College GPA with proof) from the Supportdocuments and Support_academic_records data stores. The administrator may view the file for the purpose of viewing and verifying its contents and may also export/download for the purpose of physical submission to CHED or other administrative reporting requirements.

The overall process handled by the Administrator Data Flow Diagram is through which an administrator will manage user accounts and records, administer scholarship programs, and manage scholarship applications, ranking the applicants with the assistance of Random Forest Classifier, approval of students through simulated Registrar Information System, intimating relevant information to students, and accessing and downloading students' documents and academic records if they are required.

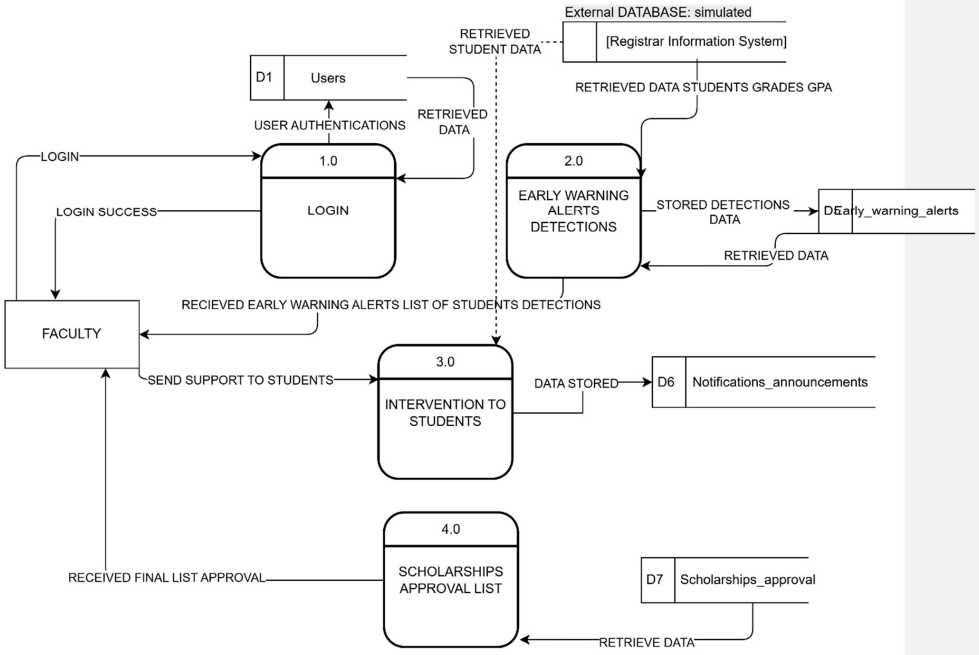


Figure 8. Faculty Data Flow Diagram

Figure 8 shows the Faculty Data Flow Diagram for Intelligent Scholarship Application, Ranking, and Early-warning Monitoring System. It shows the way in which faculty will use the system in monitoring student performances, receive and respond to early-warning notifications and academic intervention, and checking the approved scholars list. Flow of data between faculty, simulated registrar information system and data repositories is shown. The entire process begins with 1.0 Login when the faculty member logs in to the system with his valid account username and password that have been generated by the admin prior to the execution of the process. Now the faculty account is retrieved from the Users data store and check with the login credentials supplied. Once the faculty log in, they are allowed to view the 2.0 Early Warning Alerts Detection. The system uses all students' academic data including students' average

grade or GPA from the simulated Registrar Information System. The system evaluates student's academic status based on the minimum maintaining grade of 93%. Any students who has a GPA less than the minimum maintaining grade is set as academical at risk. Students' academical alert statuses are stored in the Early warning alerts data store and displayed to faculty members for follow-up action. The faculty can use the early-warning student list to see where the scholarship recipients have high-risk indicators and are likely to lose their scholarships on academic grounds. Faculty members then move on to 3.0 Intervention to Students. This would be giving advice and counseling as well as giving monitoring or whatever appropriate intervention to students whom are identified through early-warning system as in academic risk. The intervention information and announcement messages stored in the Notifications_announcements data store so that students will get notified in due time from system. Faculty can go through 4.0 Scholarships Approval List to see the ultimate approved list of scholarship holders generated by Scholarship Approval data. This ensures that faculty will view the final list of scholarship winners. It would even be possible to determine qualified students who earned the scholarship in this step.

Overall, the Faculty Data Flow Diagram illustrates the flow of a Faculty into the scholarship monitoring process: logging in to student academic records, receiving early warning of at-risk students, performing academic intervention, and accessing list of qualified scholars.

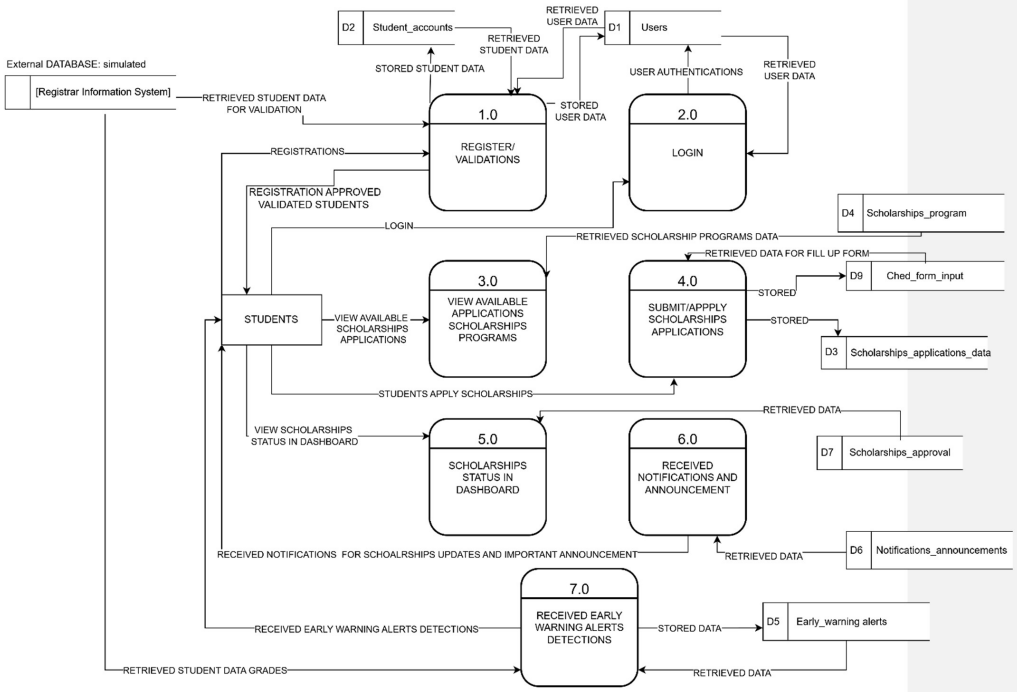


Figure 9. Student Data Flow Diagram

This Figure 9 shows the data flow diagram of Students for the Intelligent Scholarship Application, Ranking, and Early-warning monitoring system, It has been shown how a student enroll for the system, How student authenticates for the login, after that, the user log in and then fills in his profile details along with the required supporting documents along with academic transcripts and gets to know about all available scholarships, how the students can applied for the scholarships, view status of the scholarships applied by them, receive announcement and alerts and finally receive early warning related to the academic status. The flow is begun by 1.0 Register/Validation in which the user creates an account by inputting their account details. Once this is done, the user's Student ID and Full Name will be captured by the system from the simulation Registrar's Information system to check if the user's

account matches information for students registered at St. Peter's College. When the user matches against the register for valid accounts, they are considered valid. Data associated with valid accounts are then stored into the Users data store and data associated with valid accounts in the student's accounts data store are also stored into the Student_accounts data store.

After the student completes his/her successfully registration, the student will then proceed to the 2.0 Login/Profile Management/Documents Support where the system will retrieves the current student’s account in the Users data store to determine the login whether it is successful or not, if confirmed success, the student now have access into the system wherein the user can provide support to fill in personal profile and also submit his/her support documents. Documents supporting the account were the Certificate of Registration (COR), Valid ID, Signature Form, also with his/her support academic record either GWA of senior high student for freshman’s or GPA of current/alumni high student plus academic achievement evidence. Support document was save as Support_documents whereas academic records was store as Support_academic_records which serves as file references provided for administrator’s reference and to get the whole detail of support information to be export. When you log in, the student can use 3.0 View Available Scholarship Programs. All the scholarship programs available are retrieved from the Scholarships_program data store and the system lists all the scholarship programs available in front of the student. In this part you have the opportunity for anyone who would like to submit an application as students could proceed to 4.0 Submit/Apply Scholarship Applications. Whereby from this system we pull out the application parameters and detail needed. Whenever we fill in an application then the data is store in2 Schofield Schofield “Scholarships_applications_data” Data store as well as in

CHED_form_input Data store. Students can get the view of the Scholarships 5.0 Scholarship Status in the Dash Board, it is use for view and track the Scholarships applications of the students. This system uses the Scholarships_approval data store to get the scholarship applications of the particular student. In this scholarships it can display the two type of the application status like ‘Pending’ or ‘Approved’. 6.0 Received Notifications and Announcements Receiving scholarship announcement; notified notification received of events and status updates, and reminded for document submissions and intervention reports from the respective teachers and guidance staff via accessing from the data store “Notifications_announcements”. To illustrate how my system can accommodate with that in the end, I have to describe the 7.0 Received Early Warning Alerts Detection. In this step, the student’s transcript and GPA are always captured and received. If the grade of the transcript is under the required minimum standard 93%, then a early warning alert will be raised, be recorded to the store of Early_warning_alerts, the returned to the student to be seen.

To sum, the Student Data Flow Diagram showed that the enrolled student are validated by the simulated Registrar Information System, the students fill in their profile with scanned documents and student academic records, manage their applications, track the application status, send out notification & announcements to student, generate an early warning automatically according to student's academic results.

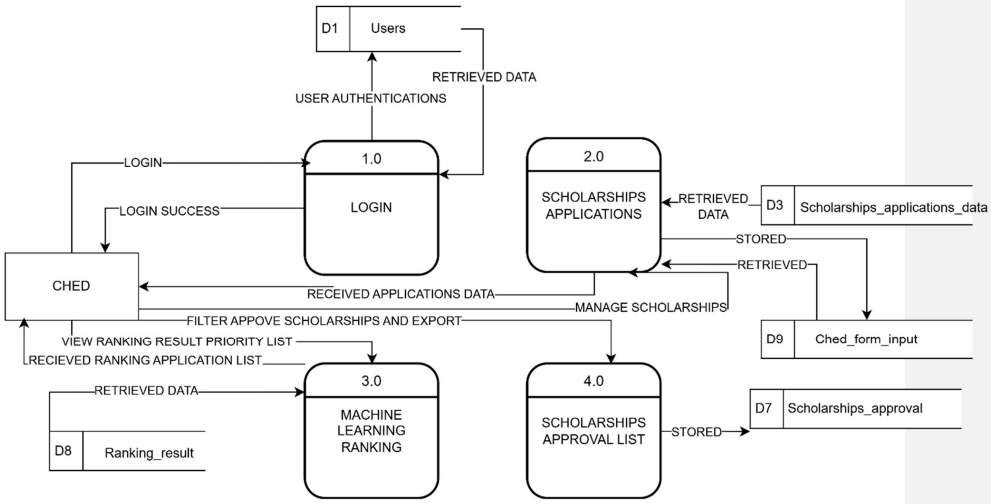


Figure 10. CHED Data Flow Diagram

Figure 10 This presents the CHED Data flow diagram for Intelligent Scholarship Application, Ranking, and Early Warning Monitoring System as in Figure 10. It shows the usage of the system by CHED staff to check the scholarship application, check the rank of scholars from machine learning rank results, pick the beneficiaries for scholarship application, and create the list of approved scholar.

1.0 Login In order for a CHED Employee to gain access to the functionalities of the system, they must perform the 1.0 Login to access the system. It involves a CHED Employee logs in the system using the allocated user account for the said employee. Login check with Users database will perform. After validation, CHED Employees can execute all scholarship functions in the system. On the first screen to appear after the staff member log in, CHED staff can go to 2.0 Scholarship Applications to be able to extract records of Scholarship Applications from the Scholarships_applications_data and manage by storing scholarship data in the CHED_form_input datastore. A CHED staff member manages the Scholarship Applications by comparing the completeness

of required data & documents of scholarship forms submitted and processing of additional documents of applicants for possible action. The CHED members can then visit to 3.0 Machine learning Ranking where the system generates the ranking of the applicants by consulting the ranking_result data store. The Ranking system is implemented using Random Forest Classifier method that ranks the candidates based on few attributes relevant for scholarship award, that primarily focuses on need to scholarship. Based on the assessment of the scholarship applications and the machine learning ranked list, CHED proceeds to 4.0 Scholarship Approval List. In this part CHED chooses the deserving scholarship applicants and approves their scholarship award. The chosen scholarship recipient's list is then stored into the Scholarships_approval data store which acts as the record for scholarship approval.

After the scholars are chosen and verified, the list of approved scholars should then be available in the system for the system administrator to check against using the simulated Registrar Information System; The administrator can check whether or not these scholars have applied or admitted into the university, he/she then also makes the necessary updates in to the Scholarship status and send out notifications, before then sending the Scholarship outcome to all the applicants.

CHED will receive the scholarship applications, use machine learning result as a supportive tool in the process of decision making, assesses the potential scholarship awardees and determines the approved recipients of the scholarship.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) represents the logical structure of the system database. It illustrates the relationships among the entities involved in the scholarship management process and defines how data are organized and

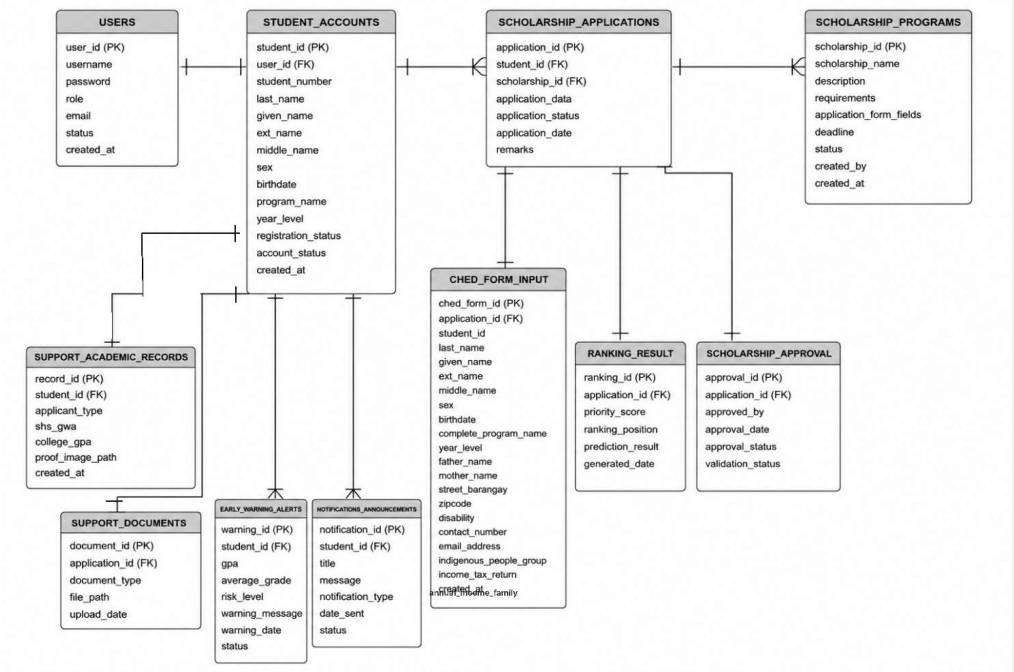


Figure 11. Entity Relationship Diagram (ERD)

As depicted by Figure 11, the ERD has entities Students, Administrators, Faculty, CHED, Scholarship programs, applications, notifications and other files that are required by the envisioned system. Figure 11 demonstrates the various relationships between these entities (using primary and foreign key relationship) in tracking data as they move from the creation of accounts, submission of application, ranking and approval and to the issuance of notifications andearly warning system (or alert). This in turn represents a well-structured and efficient system database and enables retrieval and updating of any records through the various modules while maintaining the relationship between the users and respective application and its support files.

Table 7. USERS

This depicts the structure of the Users table which contains all the account information and authentication of all users of the system like Admin, Student, Faculty

and CHED. It includes the user's unique identifier, username, encrypted password, assigned role, email address, account status and date of account creation.

Column	Data Type	Description
user_id (PK)	INT	Unique User ID
username	VARCHAR(50)	Login Username
password	VARCHAR(255)	Encrypted Password
role	ENUM('Admin','Student','Faculty','CHED')	User Role
email	VARCHAR(100)	Email Address
status	ENUM('Active','Inactive')	Account Status
created_at	DATETIME	Date Created

Table 8. STUDENT_ACCOUNTS

The following is a sample of the information collected for student accounts, including information on the student's record and reference number, student personal information (name, sex, birthdate, etc.), and academic information (program, year level, etc.).

Column	**Data Type**	**Description**
student_id (PK)	INT	Student Record ID
user_id (FK)	INT	User Reference
student_number	VARCHAR(20)	Student ID Number
last_name	VARCHAR(50)	Last Name
given_name	VARCHAR(50)	Given Name
ext_name	VARCHAR(20)	Extension Name

Column	Data Type	Description
middle_name	VARCHAR(50)	Middle Name
sex	ENUM('Male','Female')	Sex
birthdate	DATE	Birthdate
program_name	VARCHAR(150)	Program
year_level	VARCHAR(20)	Year Level
registration_status	VARCHAR(20)	Verified/Unverified
account_status	VARCHAR(20)	Status
created_at	DATETIME	Date Created

Table 9. SCHOLARSHIP_PROGRAMS

This table contains all information for scholarship programs, such as a scholarship's identifying information, description and requirements, form configuration for applications, application deadline, status, and information about who created the program and when.

Column	Data Type	Description
scholarship_id (PK)	INT	Scholarship ID
scholarship_name	VARCHAR(150)	Scholarship Name
description	TEXT	Description
requirements	TEXT	Requirements
application_form_fields	LONGTEXT	Form Fields
deadline	DATE	Deadline

Column	Data Type	Description
status	ENUM('Open','Closed')	Status
created_by	INT	Admin ID
created_at	DATETIME	Date Created

Table 10. SCHOLARSHIP_APPLICATIONS

This table contains all the information for the scholarship application, such as application identification and reference information, answers that were entered for the form, application status, application submission date and remarks.

Column	Data Type	Description
application_id (PK)	INT	Application ID
student_id (FK)	INT	Student Reference
scholarship_id (FK)	INT	Scholarship Reference
application_data	LONGTEXT	Form Answers (JSON/Text)
application_status	ENUM('Pending','Approved','Not Approved')	Status
application_date	DATETIME	Date Submitted
remarks	TEXT	Remarks

Table 11. RANKING_RESULT

This table lists all the data that is recorded for ranking the results, such as the application identification, machine learning priority score/ prediction output, resulting rank position, and date generated.

Column	Data Type	Description
ranking_id (PK)	INT	Ranking ID
application_id (FK)	INT	Application Reference
priority_score	DECIMAL(10,2)	ML Score
ranking_position	INT	Rank Position
prediction_result	VARCHAR(50)	ML Output
generated_date	DATETIME	Date Generated

Table 12. SCHOLARSHIP_APPROVAL

This table lists all of the information regarding scholarship approvals, such as the reference for the approval, the approver, the date of the approval, the final decision made and the status of the approval by the registrar.

Column	Data Type	Description
approval_id (PK)	INT	Approval ID
application_id (FK)	INT	Application Reference
approved_by	INT	CHED/Admin ID
approval_date	DATETIME	Date Approved
approval_status	ENUM('Approved')	Final Decision
validation_status	ENUM('Validated','Not Validated')	Registrar Check

Table 13. EARLY_WARNING_ALERTS

This table lists all of the information associated with an early warning alert, including a reference to the student, academic performance indicators (GPA, average

grade), the risk level associated with the alert, message for the alert, date generated, and current status of the alert.

Column	Data Type	Description
warning_id (PK)	INT	Warning ID
student_id (FK)	INT	Student Reference
gpa	DECIMAL(5,2)	GPA
average_grade	DECIMAL(5,2)	Average Grade
risk_level	VARCHAR(50)	Risk Level
warning_message	TEXT	Alert Message
warning_date	DATETIME	Date Generated
status	VARCHAR(20)	Active/Resolved

Table 14. NOTIFICATIONS_ANNOUNCEMENTS

This table contains all the data for notifications and announcements that have been sent to students including notification's reference to the student, title and message content, type, date sent and if the notification has been read.

Column	Data Type	Description
notification_id (PK)	INT	Notification ID
student_id (FK)	INT	Student Reference
title	VARCHAR(255)	Title

Column	Data Type	Description
message	TEXT	Message Content
notification_type	ENUM('Announcement','Status Update','Warning','Scholarship')	Type
date_sent	DATETIME	Date Sent
status	ENUM('Read','Unread')	Status

Table 15. CHED_FORM_INPUT

All the data collected on the CHED forms are displayed in this table, with the exception of the form's reference to the application for the scholarship, which is found on the scholarship application itself.

Column	Data Type	Description
ched_form_id (PK)	INT	Unique CHED Form ID
application_id (FK)	INT	Reference to Scholarship Application
student_id	VARCHAR(20)	Student ID
last_name	VARCHAR(50)	Last Name
given_name	VARCHAR(50)	Given Name
ext_name	VARCHAR(20)	Extension Name
middle_name	VARCHAR(50)	Middle Name
sex	ENUM('Male','Female')	Sex
birthdate	DATE	Birthdate

Column	Data Type	Description
complete_program_name	VARCHAR(150)	Complete Program Name
year_level	VARCHAR(20)	Year Level
father_name	VARCHAR(150)	Father's Full Name
mother_name	VARCHAR(150)	Mother's Full Name
street_barangay	VARCHAR(255)	Street and Barangay
zipcode	VARCHAR(10)	Zip Code
disability	VARCHAR(100)	Disability (nullable)
contact_number	VARCHAR(20)	Contact Number
email_address	VARCHAR(100)	Email Address
indigenous_people_group	VARCHAR(100)	Indigenous People Group (nullable)
income_tax_return	VARCHAR(255)	ITR file/path or details
annual_income_family	DECIMAL(10,2)	Family annual income (numeric, para sa Random Forest ranking)
created_at	DATETIME	Date Submitted

Table 16. SUPPORT_ACADEMIC_RECORDS

This table contains all of the information pertaining to the academic records (the record's reference to a student, the applicant category, details about academic performance depending on the applicant's status as a freshman or alumni, the location where the proof document was stored when it was uploaded, and the date of the upload).

Column	Data Type	Description
Column	**Data Type**	**Description**
record_id (PK)	INT	Unique Record ID
student_id (FK)	INT	Student Reference
applicant_type	ENUM('Freshman','Alumni')	Applicant Category
shs_gwa	DECIMAL(5,2)	Senior High School GWA (Freshman)
college_gpa	DECIMAL(5,2)	College GPA (Alumni)
proof_image_path	VARCHAR(255)	Storage path sa uploaded proof
created_at	DATETIME	Date Recorded
Column	**Data Type**	**Description**
record_id (PK)	INT	Unique Record ID

Table 17. SUPPORT_DOCUMENTS

This table lists all of the information recorded for supporting documents such as the document's reference to the application for scholarship, the type of document, the location where the uploaded file is stored, and when the file was uploaded.

Column	Data Type	Description
document_id (PK)	INT	Unique Document ID
application_id (FK)	INT	Reference to Scholarship Application
document_type	ENUM('COR','Valid ID','Signature Form')	Document Type

Column	Data Type	Description
file_path	VARCHAR(255)	Storage path sa uploaded file
upload_date	DATETIME	Date Uploaded

3.2 Research Locale

The study was conducted at St. Peter's college, Barangay Saray, Sabayle St., Iligan City for the Academic Year 2025-2026. The study site are the offices directly involved in scholarship administration, and monitoring of scholars such as the Office of Student Affairs and Services, the scholarship office, administration office and College of computer Studies (CCS).

At the time of the study, those aforementioned offices, that process scholarship applications, evaluation, ranking, and dissemination of results, employed predominantly manual methods which cause the delay in its administration.

The lack of a unified monitoring scheme made it difficult for interested personnel to be aware of the academic progress of scholars and therefore emphasizes the significance of the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring system introduced in Chapter 1 of the study.

3.3 Sampling Design

This part explains the sampling method used in this study. Sampling Design refers to the approach on how the people who participated in the assessment of the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System for St. Peter's College were selected. Respondents were determined as those who directly participated in the evaluation and implementation of the system.

Purposive sampling was applied, as the respondents of this study were selected because they directly utilized and evaluated the implemented system. This method was appropriate for the study since the participants had to be prior users and testers of the scholarship application and monitoring process, which is not possible using random sampling. Respondents included in the study were those who used the developed system for application, evaluation, approval, monitoring, or testing. They consisted of the student scholars, faculty members, and school administrator of St. Peter's College who used the system during the implementation and evaluation stage. These respondents were evaluated using a questionnaire anchored on the ISO/IEC 25010 Software Product Quality Model to assess the overall quality of the developed system.

To determine the required sample size, the researchers used **Slovin's Formula**, expressed as:

$$n = \frac{N}{1 + Ne^2}$$

where:

- *n* = required sample size
- *N* = total population
- *e* = margin of error (0.05 or 5%)

Given a total population of N = 50 individuals who use the system, and a margin of error of e = 0.05, the values were substituted into the formula as follows:

$$n = \frac{50}{1 + 50(0.05)^2} = \frac{50}{1 + 50(0.0025)} = \frac{50}{1 + 0.125} = \frac{50}{1.125} = 44.44$$

The outcome was rounded down to 44 respondents. Therefore out of 50 persons who were users of the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System, 44 respondents were chosen for study.

Such respondents fulfilled the study requirements and inclusion criteria. By purposive selection of respondents the data are based on real and authentic data by real and true users that participated directly to the system as all the respondents was actively participating in the system's evaluation. Those respondents that have not utilized the system at the stage of the implementation and evaluation were not included.

3.4 Respondents of the Study

This section have Forty-Four (44) student scholarship applicants, faculty representatives and school administrators of St. Peter's College who used the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System firsthand on the implementation and evaluation phase served as the respondents. Respondents were chosen from this population so that the system was assessed by those who were direct users thereof.

The participants included students to register, to be screened (selected and evaluated) for the scholarships, to track the results and receiving the notices,; the faculty to early notify alert and track their student,s academic records as well as intervene, the administrators to ensure valid application form, and to process scholarship files, record scholarship documents, ranking reports and scholarship related operations. Their response yielded very Useful data and reliable comments of use and worth of the designed system, evaluated using a questionnaire anchored on the ISO/IEC 25010 Software Product Quality Model.

Table 18.

This table shows the all respondents use for purposive sampling.

Respondent Type	**Population (N)**	**Sample Size (n)**	**Sampling Technique**
Students (Scholarship Applicants)	39	33	Purposive Sampling

Respondent Type	Population (N)	Sample Size (n)	Sampling Technique
Faculty Members	10	10	Purposive Sampling
Administrators	1	1	Purposive Sampling
Total	**50**	**44**	—

Inclusion Criteria

These are the inclusion criteria in choosing for study respondents:

- The respondents should be student, faculty or administrator at St. Peter's College.
- Must have used the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System.
- The student-respondent should have successfully applied for the scholarship.
- The faculty and the administrator's group must have utilized the academic monitoring and early-warning system
- The administrative group of respondents should have performed a certain functions related to scholarship activities and the scholarship's processes in utilizing it as well
- Must have participated in the development, testing, or evaluation of the system

Exclusion Criteria

The following individuals were excluded from the study:

- Individuals who had not used the developed system.
- Students who had not submitted scholarship applications through the system.
- Individuals who were not involved in scholarship application, evaluation, approval, or monitoring processes.
- Individuals who did not participate in the testing or evaluation of the system.
- Individuals who declined to participate in the study.

Overall, there are 44 respondents that have been actively using the developed system. These are the users that can give us some perspective on administration as well as end users.

3.5 Research Instrument

To evaluate the developed system, the study employed an adopted survey questionnaire as its sole research instrument, structured into four parts and anchored on the ISO/IEC 25010 Software Product Quality Model, consistent with the theme "Software Quality Standards Using ISO/IEC 25010" discussed in Chapter 2. Part I gathered the demographic profile of the respondents, while Part II assessed the software quality of the proposed Intelligent Scholarship Application, Ranking, and Early-Warning Monitoring System in terms of selected ISO/IEC 25010 characteristics, namely Functional Suitability, Reliability, Usability, Performance Efficiency, and Security (ISO/IEC 25010:2011). The items under each characteristic were adapted from ISO-oriented evaluation instruments used in related studies on scholarship management and automated academic systems (Ablen et al., 2025; Al-Ayyubi & Maulana, 2023; Berido et al., 2023; Remegio & Casquejo, 2025), and were operationalized to determine how well the system supports scholarship application

processing, machine learning–based applicant ranking, and early-warning monitoring of scholars within the context of St. Peter's College.

Part III consisted of open-ended questions intended to elicit respondents' comments and recommendations. The major functions and constructs evaluated in the instrument were further supported by studies on scholarship ranking and prediction (Alfina et al., 2022; Fajardo et al., 2024; Kanwal et al., 2024) and on early-warning systems (Bañeres et al., 2023; Hu et al., 2014; Plak et al., 2022), ensuring that the questionnaire comprehensively captured both the technical quality and functional relevance of the system under evaluation.

Table 19. Five-Point Likert Scale

Scale Value	Verbal Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

3.6 Validity and Reliability of the Instruments

The instrument used was adopted from software quality questions based on ISO 9126 standards, and an established Scale based on the Technology Acceptance Model (TAM). Only a few changes were made to fit the context of the current study,

such as changes in names and institutional references. No changes were made to the meaning of the items. Since the instruments adopted had gone through the validity and reliability tests, no further tests were done for this study. The results of Cronbach's alpha analysis of the data would be discussed in chapter 4.

3.7 Data Gathering Procedure

The gathering of data was done through a systematized procedure to obtain quality and credible data to evaluate the Intelligent Web Based Scholarship Application, Ranking and Early Warning Monitoring System of St. Peter's College. The following procedure was used:

- **Permission and Coordination**

  Before conducting the study, approval from the proper authorities of St. Peter's College was secured. The researchers coordinated with the concerned offices and parties involved in the system's development for the work needed.

- **Orientation of the Respondents**

  All participants were briefed about the purpose of the research, how data would be collected, and what the developed system could do. All participants were assured that their participation was voluntary, and that all collected data would be held confidential and used solely for the purpose of this research.

- **System Demonstration and Actual Usage**

  The researchers demonstrated how to use and operate the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System with the respondents. Within an ample time of around 15–30 minutes, the respondents tried out the features involving scholarship application, applicant ranking, early-warning monitoring, the notification system, and scholarship management. Based on the purposive sampling technique described in Sections 3.3 and 3.4, only respondents with actual usage of the developed system were considered.

- **Administration of the Survey Questionnaire**

After using the system, the respondents were requested to answer the survey questionnaire, which was developed based on the ISO/IEC 25010 Software Product Quality Model, to gather responses concerning the quality and impact of the developed software.

- **Collection and Verification of Responses**

  Completed questionnaires were collected and assessed for validity and completeness. Questionnaires containing blank or inaccurate responses were disregarded and excluded from the statistical analysis.

- **Data Encoding and Statistical Analysis**

  The collected data were coded into a MS-Excel and statistically treated by using suitable statistical treatment for analyzing responses of selected respondents. The quality, performance, and overall effectiveness of the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System were determined from this analysis and whether the study's objectives were met.

3.8 Scoring Procedure

The features of the software were evaluated by a 5-point Likert scale based on the information gathered by the questionnaire applied in the evaluation of the Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System for St. Peter's College. A point value was assigned to each option: 5 points for Strongly Agree, 4 points for Agree, 3 points for Neutral, 2 points for Disagree and 1 point for Strongly Disagree. The respondents' score was calculated as the average of all the aspects of software quality characteristics, weighted by the number of responses. The interpretations of the mean scores are summarised in Table 20

Table 20. Scoring Procedure and Interpretation

Scale Range	Verbal Interpretation	Description
4.21–5.00	Strongly Agree / Excellent	The respondents highly agree that the system satisfies the specified criterion and demonstrates excellent performance.
3.41–4.20	Agree / Very Satisfactory	The respondents agree that the system satisfies the specified criterion with very satisfactory performance.
2.61–3.40	Neutral / Satisfactory	The respondents are moderately satisfied or undecided regarding the system's performance.
1.81–2.60	Disagree / Fair	The respondents disagree that the system satisfies the specified criterion satisfactorily.
1.00–1.80	Strongly Disagree / Poor	The respondents strongly disagree, indicating that the system does not satisfy the specified criterion.

The same scale ranges will be used in Chapter 4 for the average scores that are obtained for each of the selected qualities and characteristics of ISO/IEC 25010 to determine the overall quality and performance of the software.

3.9 Statistical Treatment of Data

This part shows and explains the statistical tools used to analyse the data collected from respondents to evaluate St. Peter's College's Intelligent Web-Based Scholarship Application, Ranking, and Early-Warning Monitoring System, summarises

the results, and examines the overall and detailed evaluation of the system in relation to the specific criteria for ISO/IEC 25010 Software Quality Model.

Frequency and percentage were used to interpret the distribution of respondents by classification of the users which includes student scholarship applicants, faculty members and directors of the school. Frequency is the number of people responding to a group and percentage is the proportion of people in a group relative to the total. The percentage was computed using the formula:

$$P = f/N \times 100$$

Where:

- P = Percentage
- f = Frequency of responses
- N = Total number of respondents

These statistical measures were used to present the demographic distribution of the respondents included in the study.

The average score of all the respondents was calculated for each of the five-point Likert scale assessment items, using a weighted mean. In this study, the five software quality characteristics of developed system including functional suitability, reliability, usability, performance efficiency and security, the weighted mean is considered the most appropriate statistic for evaluating the developed system against each of the ISO/IEC 25010 software quality characteristics. The weighted mean was computed using the formula: $\bar{x} = (\sum fx)/N$

Where:

- $\bar{x}$ = Weighted Mean
- f = Frequency of responses
- x = Scale value of each response
- N = Total number of respondents

The computed weighted mean served as the basis for interpreting the respondents' overall evaluation of each software quality characteristic.

The standard deviation was computed to indicate the spread or dispersion of the respondents' answer around the mean value obtained. Low standard deviation means that the answers were near the mean value and high standard deviation means that the answers were widely distributed from the mean value.

Data obtained was processed, coded, tabulated and analyzed using Microsoft Excel software and the following statistical data present and analyze was calculated and presented in chapter 4 such as Mean, Standard deviation and frequency percentage and weighted mean.

3.10 Ethical Considerations

The ethical considerations have been taken into account while conducting this research in order to ensure the protection of the rights and well-being of the participants who took part in the evaluation of the Intelligent Web-based Scholarship Application, Ranking and Early Warning Monitoring System at St. Peter's College.

Before the process of collecting data commenced, the subjects were enlightened on the objective, aim, process, importance, and expected results of the research. Informed consent was gotten from all subjects before they participated in the system evaluation. It was made clear to the subjects that participation in the research was voluntary and that they could stop at any time without repercussions.

Confidentiality and privacy of the respondents were strictly ensured during the course of conducting the research. The personal information of the respondents was kept in confidence, and no reports, articles, or presentations had been published or shared by disclosing any respondent's identity. The respondent's names were replaced by codes/identification numbers, and data were provided only in aggregate form.

The research study abided by the requirements set out in the Data Privacy Act of 2012, also referred to as Republic Act No. 10173. All the data gathered from the respondents was used strictly for academic and research purposes. The data collected was stored in a password-protected computer document which was only available to the researchers. Proper measures were taken to ensure that all data collected was confidential and safe during the whole research process.

Before the execution of the research study, permission to carry out the research was obtained from the relevant bodies at St. Peter's College. The data collection and system evaluation process could not begin until institutional permission had been obtained.

Proper referencing standards according to accepted practice were employed in dealing with all the theories, concepts, literature and other reading materials of this research. It was attempted to carry out this research in a responsible manner by avoiding plagiarism and giving adequate credits to other authors for intellectual work involved.